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Mesozoic litho- and magneto-stratigraphic evidence from the central Tibetan Plateau for megamonsoon evolution and potential evaporites

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ABSTRACT

The megamonsoon was a striking event that profoundly impacted the climatic environment and related mineral sources (salts, coals and oil-gases) in the Mesozoic. How this event impacted Asia is unknown. Here, we firstly reported a Mesozoic stratigraphic sequence in the northern Qiangtang Basin, in the central Tibetan Plateau, based on lithofacies and chronologies of paleontology and magnetostratigraphy. How the planetary and megamonsoon circulations controlled the Asian climate with time has been recorded. Using the basic principles of physical geography, present analogs and a newly developed model, the evolution of the stratigraphic sequence was interpreted to demonstrate that the Qiangtang Basin has been subjected to a megamonsoon climate with heavy precipitation during its northward movement since the Latest Permian but has been subjected to a dry environment due to moving into the northern hemisphere subtropic high zone in the Middle Triassic and monsoonal retreat in the early Late Jurassic (early Oxfordian) approximately 161 Ma. The coupling of the hot-dry climate and the close of the Meso-Tethys, along with sea retreat in the Late Jurassic, ensured a large potential time window for evaporite (potash) formation in the Qiangtang Basin, while the tropic megamonsoon rain forest in the Late Permian to the Early Triassic and in the Late Triassic to Middle Jurassic favored the formation of coal and hydrocarbon source rocks.

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1. Introduction

The bases of China mostly comprise blocks/terranes of micro to small scales (Fig. 1a) that came in different times from the eastern margin of the Gondwana super-continent, which was located in the southern hemisphere during the Paleozoic and Mesozoic. The northward drifts of these blocks and their collisions with Paleo-Asia constituted the major geologic history of China (e.g., Chang et al., 1986; Zhao et al., 1996; Zhu et al., 1998; Yin and Harrison, 2000; Yang and Besse, 2001; Xiao et al., 2005; Metcalfe, 2013; Zhai, 2013; Zhang et al., 2015) and determined the mostly regional paleoclimatic environment patterns during these periods (Boucot et al., 2009; Qiang and Shen, 2014; Metcalfe et al., 2015). Northward drifting of the Qiangtang and Lhasa blocks and closures of the Paleo- and Meso-Tethys Oceans (Sengör, 1987; Huang et al., 1992; Chang et al., 1995; Zhu et al., 2011; Metcalfe, 2013; Zhu et al., 2013), which now are the main parts of the Tibetan Plateau (Yin and Harrison, 2000; Wang et al., 2014), are representative geologic events. Concurrent with these geologic events was the development and vanishing of the megamonsoon in the Mesozoic (Kutzbach and Gallimore, 1989; Parrish, 1993; Weissert and Mohr, 1996; Rais et al., 2007). The majority of works have focused on the histories, features and mechanisms of the drifting, collision, orogeny and related mineral source formation of the blocks in China (e.g., Zhang et al., 1995; Zhu et al., 1998; Huang et al., 1999; Yin and Harrison, 2000; Xiao et al., 2005; Hou et al., 2007; Zhai, 2013; Zaw et al., 2014; Zhang et al., 2015), but almost no work has addressed the pattern and mechanism of the related climatic changes in relation to the Mesozoic megamonsoon. Here, we used the principles of physical geography, the general patterns and features of modern climate (present analogs) that are determined by interactions between modern air masses and continents and orogenic mountains of various sizes and positions, and knowledge from numeric modeling to build an air mass-circulation-climate working model based on insolation–continent–ocean interaction. Then, used this working model to interpret paleoclimatic changes and patterns recorded in the Mesozoic stratigraphy, for which we knew only the relative climatic characteristics (hot to cold and humid

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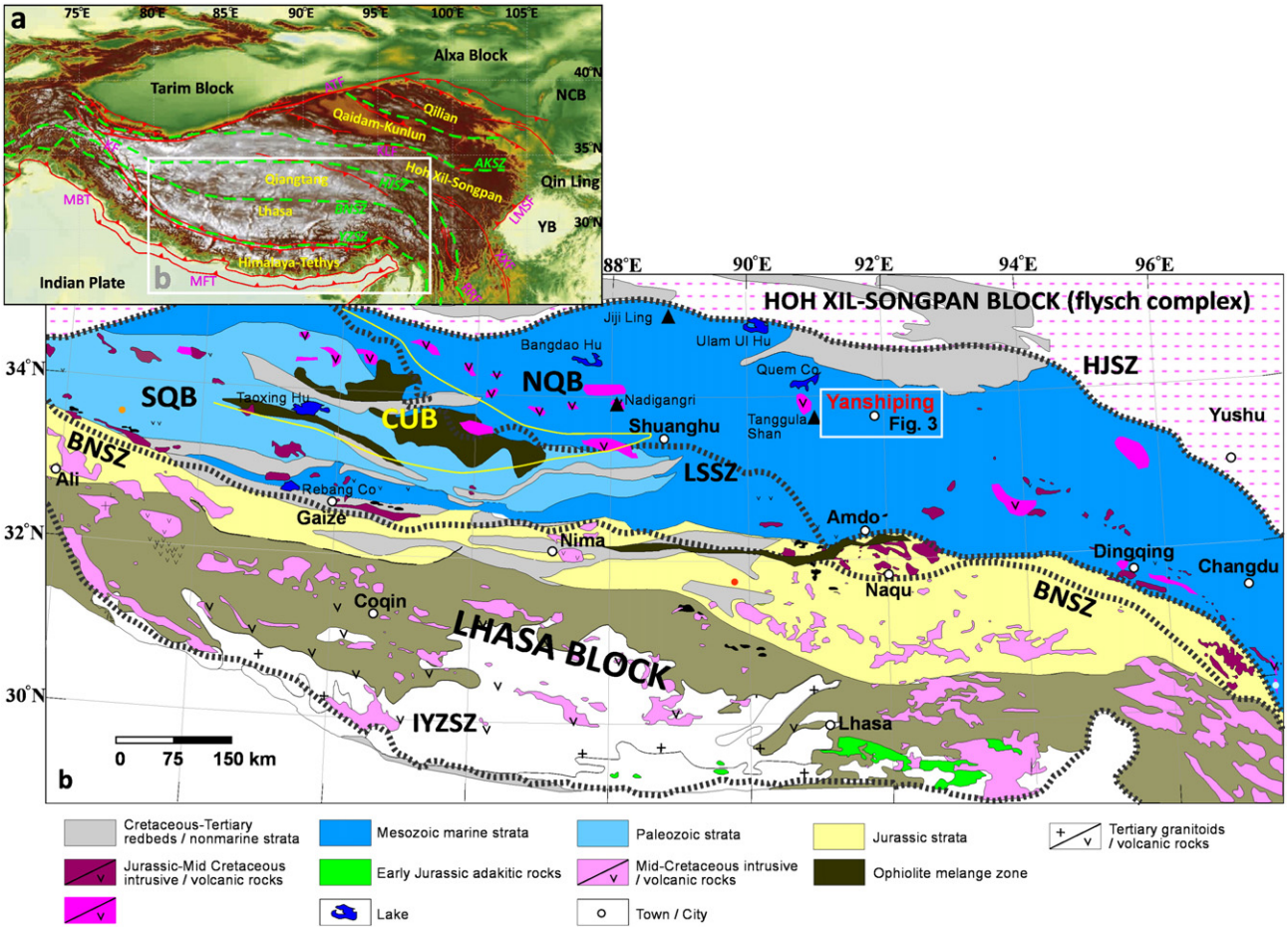


Fig. 1. Summarized tectonosedimentary map of the Qiangtang Basin showing the tectonic and stratigraphic frames of the basin and the location of the studied area (compiled from Kapp et al., 2003, 2007). Inset map showing the blocks of China and the Tibetan Plateau. Detailed ages of the intrusive and volcanic rocks refer to Yan et al. (2016). AKSZ: Anemaqen–Kunlun Suture Zone; HJSZ: Hoh Xil–Jinsha Suture Zone; BNSZ: Bangong–Nujiang Suture Zone; IYSZ: Indus–Yalung Zangpo Suture Zone; CUB: Central Uplift Belt; NCB: North China Block; YB: Yangtze Block; ATF: Altyn Tagh Fault; KLF: Kunlun Fault; XXF: Xianshuihe–Xiaojiang Fault; RRF: Red River Fault; LMSF: Longmen Shan Fault; KF: Karakoram Fault; MBT: Main Boundary Fault; and MFT: Main Frontal Fault.

to dry) without clear knowledge of the spatial relationships and mechanisms. With this approach, we put the Mesozoic paleoclimatic change of the Tibetan Plateau in logical order, both temporally and spatially, and revealed their driving mechanisms and relationships with the megamonsoon. We first applied combined paleontological and paleomagnetic studies to constrain the ages of the Mesozoic stratigraphy in the Qiangtang Block, then obtained the paleoclimatic records from lithofacies analyses, and finally integrated with other related world records and interpreted their spatial–temporal evolution and driving mechanisms using the working model.

2. Geologic setting

The Tibetan Plateau is assembled by a series of small E–W stretched blocks/terranes southwards, including the Qilian, Qaidam-Kunlun, Hoh Xil-Songpan, Qiangtang and Lhasa blocks and orogenic belts (former suture zones) (Fig. 1). The Qiangtang Block is approximately 1600 × 300 km (40 × 10^4 km^2), the largest block/basin in the Tibetan Plateau (Mesozoic basin at ~18.4 × 10^4 km^2). It is bounded by the Hoh Xil–Jinsha Suture Zone (HJSZ) to the north and the Bangong–Nujiang Suture Zone (BNSZ) to the south, which separate the block from the Hoh Xil–Songpan and Lhasa blocks, respectively (Chang et al., 1986; Yin and Harrison, 2000; Kapp et al., 2003, 2007; Wang et al., 2014; Fig. 1). Recent studies show that it is actually constructed by the North (or eastern) Qiangtang Block (NQB) and South (or western) Qiangtang Block (SQB), which are separated by the NWW-striking Longmu Co–Shuanghu Suture Zone (LSSZ) through the central Qiangtang. The region now is presented as the Central Uplift Belt (CUB) and consists of blueschist, eclogite, metabasaltic rocks, ophiolitic mélange, basalt, metapelite, marble, minor chert, and ultramafic rocks, indicating ocean-floor and sea mount rock associations (Zhai et al., 2011 and references therein; Metcalfe, 2013) and four phases of volcanic rocks representing different tectonic environments (Fig. 1). There are some basalt beds (e.g., BGMX, 1993; Li et al., 1995, 2009) and voluminous E–W mafic dikes that were dated at 302 and 284 Ma (Zhai et al., 2009) in the SQT, probably produced by early rifting of the QT away from the Gondwana (Zhu et al., 2013) (Fig. 1b); dike basalt and andesite of 237–232 Ma distributed along deep faults (Fig. 1b), indicating intraplate extension environment; U–Pb zircon ages of 232–205 Ma of intermediate-felsic intrusive rocks as splendid Tanggula Shan batholith and their corresponding effusive rocks (Fig. 1b), reflecting oceanic plate subduction and arc, intraplate extension and exhumation; and Ar–Ar ages of 156–130 Ma of dike and effusive basalt and andesite showing successive extensional environments (Jia et al., 2006; Wang et al., 2007a; Zhang et al., 2012; Metcalfe, 2013; Peng et al., 2015) (Fig. 1b). The first phase of the volcanic rocks, together with the recently obtained U–Pb zircon ages of 230 to 237 Ma of the blueschists and eclogites (Zhai et al., 2011) and the occurrence of the Middle Triassic molasse in the LSSZ demonstrated that the LSSZ was formed in the Middle Triassic (Yan et al., 2016). It has recently been thought to represent the closure of the main Paleo-Tethys Ocean (Metcalfe, 2013; Zhang et al., 2013).

The basement of the QB is thought to have a Gondwana origin. Paleomagnetism, paleontology and lithostratigraphy have shown that the QB was disassembled from the eastern margin of the Gondwana continent in the Early Permian (Zhu et al., 2013) and submerged in the sea as an epicontinental sea shelf. Then, it drifted fast northwards and collided with the Hoh Xil-Songpan Block in the Late Triassic, forming a syntectonic foreland basin and thrust-fold system along the southern margin of the HJSZ and the northern margin of the QB (Li et al., 2002). The successive Lhasa Block collided obliquely westwards with the QB from the Late Jurassic to the early Cretaceous, forming a syntectonic foreland basin and thrust-fold system along the southern margin of the BNSZ and giving rise to a westward retreat and closure of the Meso-Tethys Sea, ending the Jurassic epicontinental sea on the QB and uplifting it as an erosion and provenance region (Sengör, 1987; Li et al., 2002). Subsequent subduction of the Neo-Tethys oceanic plate and collisions of the Neo-Tethys arcs and India with the Lhasa Block did not change the tectonic configuration of the QB, but formed some small intrablock (intermontane) basins filled with continental red beds along former faults and greatly deformed and raised the QB to the present height (Fig. 1b).

The QB now occupies the central part of the Tibetan Plateau and has an average elevation of more than 5000 m (thus called the Qiangtang Plateau) and a dry-cold climate (mean annual precipitation and temperature: 50–300 and −1 to −6 °C, respectively) controlled by interaction of the Asian monsoon and the westerlies. Desert steppe to desert landscapes predominate the plateau. Small ice caps and glaciers occur on peaks of high mountains standing on the plateau and supply melt water to numerous small brackish and salt lakes on the plateau.

3. Stratigraphy and sampling section

3.1. Stratigraphic sequence of the basin

The Qiangtang Block (Basin) is mostly composed of three sets of stratigraphy. The pre-Devonian (pre-Cambrian?) basement, late Paleozoic–Mesozoic (D–J) marine sediments and late Mesozoic–Cenozoic (K–Q) intermontane continental red bed sediments. All three sets of the stratigraphy are intercalated with some volcanic rocks representing different tectonic environments (Figs. 1 and 2).

The basement of the Qiangtang Block is uncertain because it is covered mostly by thick Mesozoic stratigraphy. The exposed epimetamorphic rock series of slate, phyllite and epimetamorphic feldspathic quartz sandstone and quartz sandstone intercalated with crystalline limestone of Amugang, Gemuri and Mayigangri Groups of pre-Devonian (Precambrian?) period

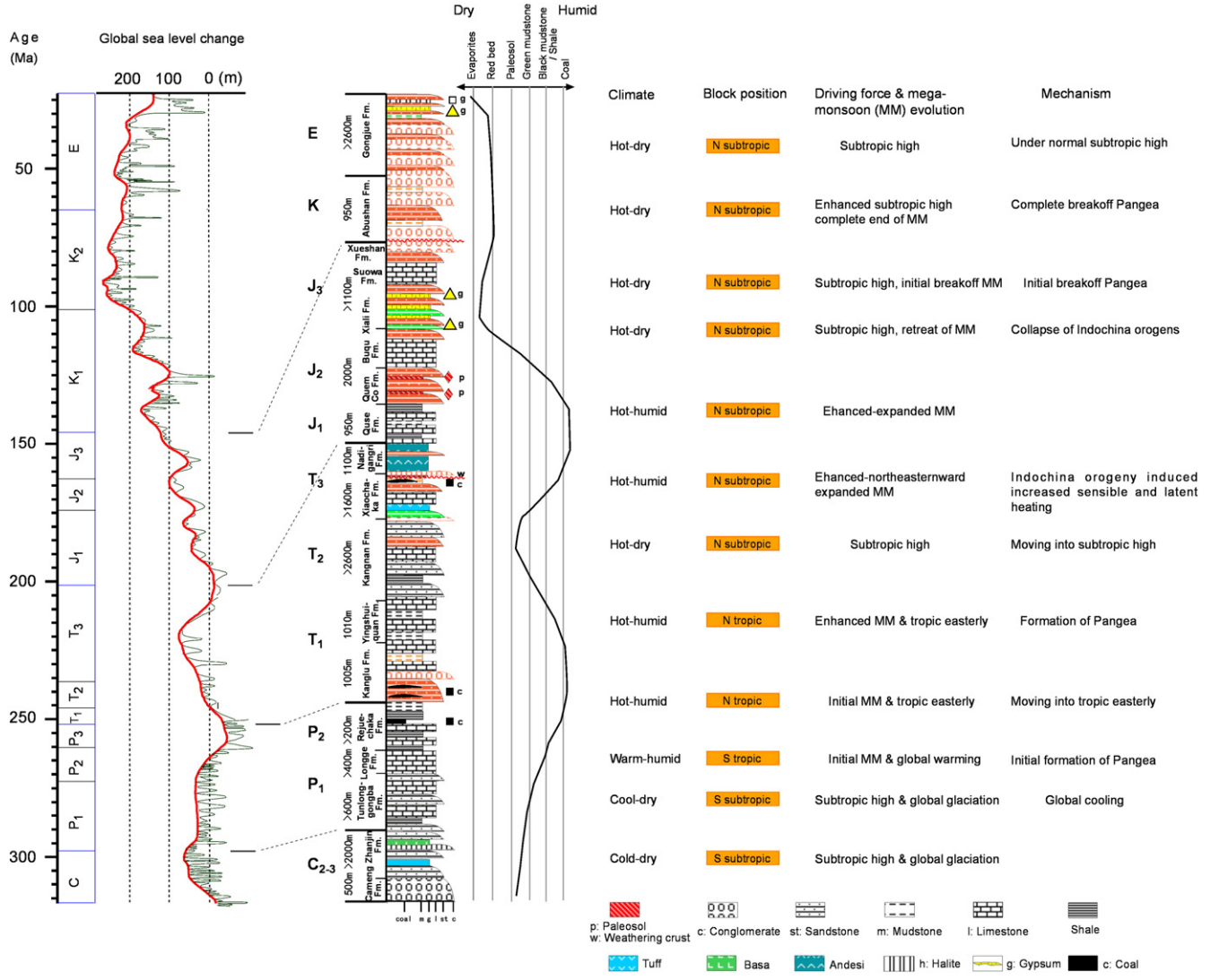


Fig. 2. Stratigraphic sequence and lithology of the late Paleozoic to Mesozoic in the Qiangtang Basin and its recorded climatic change and megamonsoon evolution with possible mechanisms.
Global sea level change was taken from Haq and Schutter (2008).

in the LSSZ have been used to estimate the basement composition of the NQB (Wang et al., 1987; Li et al., 2005; Zhai et al., 2011).

The late Paleozoic–Mesozoic (D–J) stratigraphy is unconformably superimposed on the basement. The Devonian to Permian sediments are mostly shallow sea carbonates formed in a continent passive margin, while the Mesozoic sediments are mostly shallow sea to sea-land interbedding carbonates and siliciclastic sediments, recording two cycles of tectonic evolution from a pre-collisional passive margin basin through a syn-collisional foreland basin to a post-collisional extensional depression basin. The late Mesozoic–Cenozoic sediments indicate mostly intra-continental deformation in response to compressions from the subduction of the Meso-Tethys Sea and subsequent indentations of the Meso-Tethys arcs and Indian Plate into the Paleo-Asia Plate (Fig. 2).

In sequence, the early Paleozoic rocks are a stable shallow marine carbonate platform of sandstones, mudstones and limestones of Late Ordovician Yinshuihe Fm. and Silurian Puercuo Fm. and are exposed in only a very limited area (tens of km^2) to the south of Mayigangri (Mt.) in the LSSZ or CUB (see Fig. 1 for location) (Hao et al., 1989; Li et al., 2005). The carbonate sequences continued to the Devonian, Carboniferous and Middle Permian. The Carboniferous rocks are exposed in only a few sites along the CUB and are characterized by low-maturity clastic rocks, intercalated with basic volcanic rocks of possibly continental rift and aquatillites of miscellaneous conglomerates and cold water fauna (only for the Late Carboniferous of the SQB) (Hao et al., 1999; Li et al., 2005) (Fig. 1b). The Early Permian sediments mimic the Late Carboniferous, containing cold water fauna and tills in the SQB and shallow marine carbonate formation of limestones and sandstones for the whole Qiangtang Basin. However, the Late Permian sediments changed to marine-terrigenous interbedding fascies of limestones containing warm water fauna, sandstones and mudstones intercalated with several layers of coals bearing famous Gigantopteris flora (Hao et al., 1989; Zhang et al., 2013) (Fig. 2). In summary, the Devonian to Permian stratigraphy indicates a stable marine tectonic–depositional environment of continent passive margin having a Gondwana origin (for $D\text{-}P_1$).

The Triassic is special and diversiform in lithostratigraphy due to closure of the Paleo-Tethys oceans (LSSZ and HJSZ) during the Indochina movement. The Early and Middle Triassic sedimentation occurs only in the NQB, CUB and HJSZ (BGMRXAR,1993; Hao et al., 1989; He et al., 2002). It is generally characterized by two intervals of siliciclastic and carbonate rocks in the NQB. The exposed Early and Middle Triassic sediments in the southern margin of the NQB and the CUB are upwards purple-brown red conglomerates to mudstones intercalated with some coals in the river delta plain and coast environments (Kanglu Fm., Induan Stage by fossil), muddy and oolitic limestones and marl of shallow sea carbonate platform (Yingshuiquan Fm., Olenekian Stage by fossil), and molasse greenish–grayish sandy shale and very thin muddy limestone of shallow seas (Kangnan Fm., Anisian Stage by fossil) (BGMRXAR,1993; Hao et al., 1989) (Fig. 2). The closer to the northern margin of the CUB, the coarser the Kangnan Fm. becomes and is dominated by greenish–grayish sandstones intercalated with some conglomerates, siltstones–mudstones and marls of continental margin facies (Hao et al., 1989; He et al., 2002). In the Tumengela depression in the southern margin of the eastern CUB, the Kangnan Fm. changes to fine purple sandstone, siltstone and mudstone intercalated with thin-layer gypsum and marl of the tidal-flat and lagoon facies (He et al., 2002) (Fig. 2). In the northern margin of the NQB, only the Middle Triassic limestone of the carbonate platform containing typical Tethys bivalves of *Asoella* sp., *Chlamys* cf. *mysica* (Bittner) and *Leptochodria* sp. is exposed near Jiji Ling (Fig. 2 and see Fig. 1 for location) (He et al., 2002). However, in the HJSZ to the north of Jiji Ling, very thick (>3000 m) flysch sandstones and mudstones intercalated with the Early–Middle Triassic andesitic volcanic rocks occur (Hao et al., 1999; Fang and Li, 2005) (Fig. 2).

The Late Triassic sediments distribute widely over the whole QB, superimposing unconformably on the former eroded regions (SQB and eastern NQB) and the Early–Middle Triassic around the NQB margins but are continuous from the Early–Middle Triassic in the central NQB (revealed by oil-gas boreholes and seismostratigraphy). In the northern margin and northeastern and eastern parts of the NQB, the Late Triassic sediments present as three distinct intervals: the lower molasse of purple conglomerates and sandstones intercalated with some gypsum of the coast river delta (Jiapila Fm., Carnian Stage by fossils) reflecting closure of the HJSZ and major collision of the QT with the HSB, the middle grayish limestone of the carbonate platform intercalated with some volcanic rocks (Bolila Fm., late Carnian–early Norian Stages by fossils), and the upper greenish–grayish sandstones to black sandy mudstones intercalated with coals of sea–land interbedding facies (shelf, near shore, tidal-plate, river delta, and coast marshes) (Bagong Fm., late Norian to Rhaetian Stages by fossils), indicating a post-collision extensional environment. It is rich in hot-humid types of pollen-spores of *Cyathidites* sp., *Deltoidospora magna*, *Dictyophyllidites* sp., *Concarisporites* sp., *Stereisporites* sp., *Chasmatosporites minor*, *Limatulasporites* sp., *Klulisporites* sp., *Cycadopites minimus*, being common components of the Norian–Rhaetian Stages (He et al., 2002; Fang and Li, 2005). In the remaining parts of the QB, the Late Triassic sediments are mostly limestones (lower) and grayish sandstones to black shales intercalated with marls and limestones (middle–upper) of carbonate–clastic formations of deep to shallow shelf (Xiaochaka Fm. in the NQB or Rigancuopei Group in the SQB) (BGMX, 1982; Hao et al., 1989; He et al., 2002; Fang and Li, 2005). Around the northern and southern margins of the CUB and in the NW and NE NQB, basic to andesitic volcanic and pyroclastic rocks intercalated with some conglomerates and sandstones, called Xiaochaka volcanic rocks and Nadigangri Fm., occurred, respectively, below (only near Xiaochaka–Shuanghu) and above the Xiaochaka Fm. (He et al., 2002; Fang and Li, 2005) (Figs. 1 and 2). SHRIMP U–Pb dating of zircon grains yielded ages of 216–205 Ma for the Nadigangri Fm. (Wang et al., 2007a) (Fig. 2), but no precise dating for the Xiaochaka volcanic rocks. Both volcanic rocks indicate a post-collision extensional environment in response to collisions of the SQB with NQB and the NQB with HSB (Yan et al., 2016). Deep weathering laterite crust is found on the top of the Xiaochaka Fm. at Xiaochaka and Juhuashan and Shishuihe near the southern and northern margins of the western CUB, indicating a hiatus and warm-humid climate after the Xiaochaka Fm. (Wang et al., 2007b).

The Jurassic stratigraphy covers the whole Qiangtang Basin, except for the Early Jurassic, when the NQS did not receive sedimentation but only some volcanic rocks (Hao et al., 1999; Fang and Li, 2005) (Fig. 1). Thus, the QB is sometimes called the Jurassic basin. The Middle–Late Jurassic environment was quite stable and uniform in the basin and was characterized by limestones of the shallow sea carbonate platform in the NQB center and SQB and land–sea interbedding siliciclasts (mostly purple-red sandstones to mudstones intercalated with some gypsum or coals) and limestones, represented by the Yanshiping Group in the NQB margins (Hao et al., 1989; Fang and Li, 2005) (Fig. 2), indicating a regional depression in continuous response to the post-collision collapse of the Late Triassic orogeny of the HJSZ and the break-off of the Meso-Tethys ocean plate beneath the QT (Metcalfe, 2013; Yan et al., 2016).

After the Jurassic, the Cretaceous to Cenozoic stratigraphy are distributed limitedly in the QB and only in some intrablock narrow basins along deep faults and are mostly continental red beds of conglomerates to mudstones intercalated with some coals or gypsum (Hao et al., 1999; Fang and Li, 2005) (Fig. 3), suggesting an overall upheaval of the QB. The stratigraphy is subject to strong faulting and folding, indicating the effects of the Neo-Tethys closure and collision of India with Asia.

3.2. Measured section and sampling

We measured and paleomagnetically sampled the Yanshiping (YSP)-type section of the Jurassic, the YSP Group, at Yanshiping on the northern slope of the Tanggula Shan (Mts.) (eastern CUB) in the eastern NQT (33°33′19.9″N–33°35′11.3″N, 92°01′57.3″E–92°03′44.3″E) (Fig. 3), where the middle–upper Jurassic strata are well exposed. The YSP

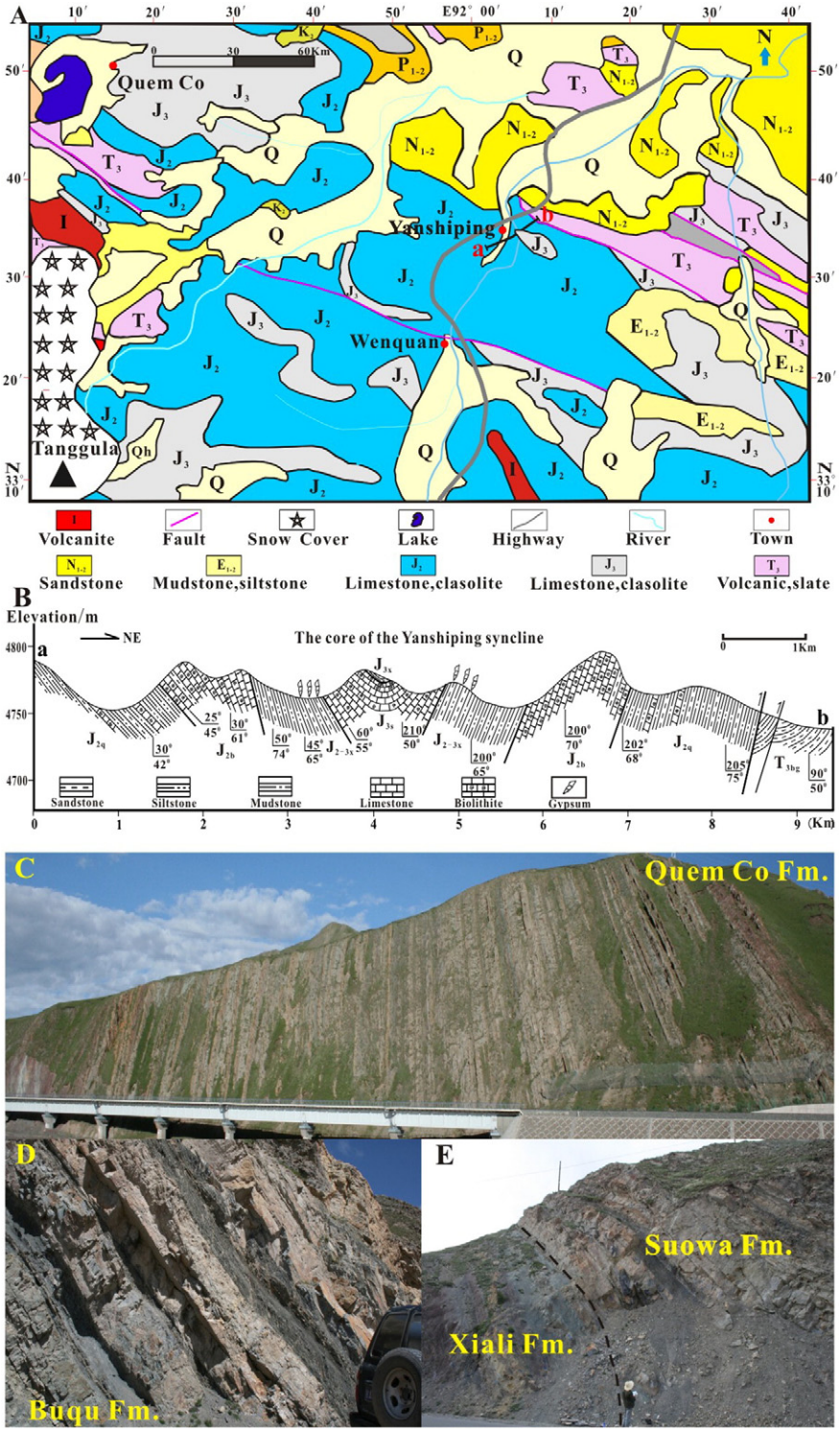


Fig. 3. (A) Geologic map of the YSP area showing the location of the YSP cross-section (a–b) in (B); Field shots of the Quem Co (C), Buqu (D) and Xiali–Suowa (E) Fms.

section is approximately 3300 m thick (sampled 3120 m) and consists of the YSP Group that is upwards subdivided into the Quem Co (J_{2q}), Buqu (J_{2b}), Xiali ($J_{2\text{-}3X}$), Suowa (J_{3s}) and Xueshan (J_{3x}) Formations (BGMX, 1982; Hao et al., 1999; Fig. 3), with a conformable contact between each two adjacent formations. The remarkable feature of the five formations is characterized by alternations of dominant purple-red sandstones to grayish limestones of land–sea interbedding facies (Fig. 3; see also Fig. 11).

The Quem Co Fm. consists of alternating purple sandstones to mudstones with associated paleosols of the delta plain (Figs. 2 and 3C; see also Fig. 11). The paleosols are luvic cambisols to luvisols (drab to brown soils) characterized by distinct Bt horizons with or without a striking Bk horizon (Fig. 4a–d). The Bt horizon is deep purple to red and silty clay-textured and has a well-developed fine granular pedostructure, abundant distinct shining purple-red ferruginous clay coatings and some large root channel silt-clay infillings (Fig. 4b). The Bk horizon is manifested as mixed colors of white, gray to grayish purple, some to many carbonate nodules 0.5–2 cm diameter, with coatings and root channel infillings (Fig. 4c, d). Abundant fresh water and marine-brackish bivalves were found in the sandstones, respectively,

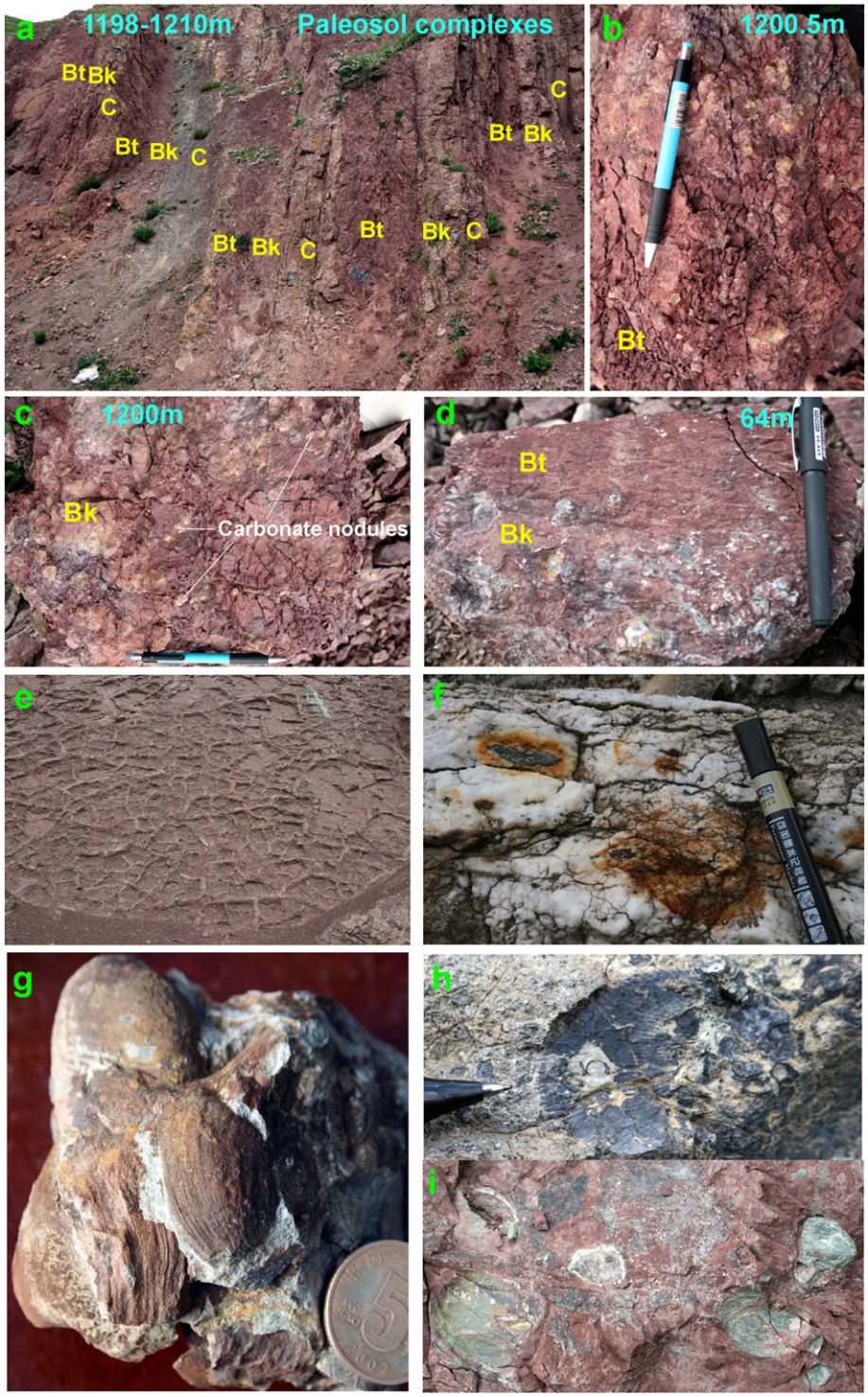


Fig. 4. Representative field photos showing paleosols, sedimentary features and fossils in the YSP section. a) Occurrence and horizons of paleosols (luvic cambisols–luvisols) in the section heights of 1198–1210 m, Quem Co Fm.; b) pedofeatures of the luvic Bt horizon showing well-developed granular ped structure and red shining iron clay coatings on peds, upper Quem Co Fm.; c) pedofeatures of the calcic Bk horizon showing down-leached carbonate precipitates as grayish carbonate nodules ca. 0.5 cm, upper Quem Co Fm.; d) pedofeatures of hand samples showing clear Bt–Bk horizons (note the good pedostructure in Bt and the white carbonate nodules in Bk), lower Quem Co Fm.; e) Mud cracks, Xiali Fm.; f) Thinly bedded alabastrine gypsum, Xiali Fm.; g) fresh water bivalves of *Lamprotula* (*Eolamprotula*) sp. nov., lower Quem Co Fm.; h) marine bivalves of *Pseudolimea duplicata* (J. de C. Sowerby), Buqu Fm.; and i) marine bivalves of *Anisocardia* (*Anisocardia*) cf. *inversa* (Lycett), Xiali Fm.

in the lower and upper parts of the Quem Co Fm., suggesting ages from the Bajocian to Bathonian Stages (Table 1 and Fig. 4 g).

The Buqu Fm. is thick white-grayish massive to laminated and bioclastic limestones intercalated with some black shale of the open sea carbonate platform (Fig. 3D; see also Fig. 11) containing abundant typical shallow marine bivalves of the Bathonian–Callovian Stages (Table 1 and Fig. 4 h).

The Xiali Fm. comprises purple-red and yellow-green sandstones, siltstones, and multicolored (dark red, gray, grayish and light yellow) mudstones intercalated with thin, dark gray beds of microcrystalline

Table 1
Fossil bivalves found in the Yanshiping section and their constrained biostratigraphy.

Stratigraphy	Fossil bivalves	Ecological environment	Age	Reference
Xueshan Fm.	*Peregrinoconcha yunanensis*, *P. perlonga* *Myopholas multicostata* (Agassiz), *Pleuromya* sp., *Sinonaia* sp.	Marine-brackish	Tithonian Kimmeridgian	Bai (1989)
Upper Suowa Fm.	Assemblage *Entolium–Myopholas*: *Entolium corneolum* (Young et Bird), *Astrarte* (*Astarte*) *nummus* (Sauvage), and *Myopholas multicostata* (Agassiz), *Myopholas* sp., *M. percostata* Douville, *Plagiostrma* cf. *rigidus* (Sowerby).	Marine	Oxfordian-Kimmeridgian	This study; Bai, 1989
Lower Suowa Fm.	Assemblage *Gervillella aviculoides–Radulopecten fibrosus*: *Gervillella aviculoides* (Sowerby), *Lophagregarea* (Sowerby), *Isognomon mytiloides*, *Modiolus bipartitus* Sowerby, *Corbicellopsis laevis* (Sowerby), *Myopholas douvillei*, *Pteroperna* cf. *polydon* (Dunker), *Radulopecten fibrosus*, *Chlamy* (*Radulopecten*) *midas* (d'Orbigny), *Mytilus angulatus* (Young et Bird). *Gervillella aviculoides* (Sowerby), *Modiolus* sp., *Pteroperna* cf. *polyodom* (Dunker), Arca sp., *Protocardia* (*Protocardia*) *purbeckensis* (Loriol et Jaccard), *Astarde cf. cordala* Trauschold; *Myopholas multicostata* (Agassiz), *M. douvillei* Lissajous, *Pteroperna cf. polyodom* (Dunker), *Radulopecten cf. scarburgensis* (Young et Bird), *R. fibrosus* (Sowerby), *Chlamys* (*Chlamys*) *tentoria* (Schlotheim), *G. laevis* Wen, *Anisocardia* sp., *Pseudolimea duplicate* (Sowerby), *Platymyoidea? Vaulingnyacerisis* (Loriol), *Ostrea* sp. *Entolium* sp., *Anisocardia* (*Aisocardia*) ex gr. *islipensis* (Lycett), *Mactromya*?sp., *Actinostreon gregareum* (J. Sowerby), *Ceratomya* sp., gen. et sp. indet., *Mactromya*?sp.	Marine	Oxfordian	Chen et al., 2005 Bai, 1989 This study
Xiali Fm.	Assemblage *Anisocardia* ex gr. *Islipensis/tenera–Modiala biparta–Isognomon cf. Bergeroni:* *Anisocardia* (*Aisocardia*) ex gr. *islipensis* (Lycett), *A.* (*A.*) cf. *islipensis* (Lycett), *Actinostreon gregareum* (J. Sowerby), *Gervillella qinghaiensis* Wen., *Ceratomya* sp., *Placunopsis* sp., *Corbula* sp., *Pleuromya uniformis* (J. Sowerby), *Quenstedtia* sp., *Thracia* sp., *Tancredia* sp., *Pholadomya qinghaiensis* Wen, *Pinna*? sp. *Plagiostoma* cf. *rigida* (Sowerby), *Chlamys* (*Chlamys*) *splendens* (Dollfus), *Pleroperna burensis* Loriol, *Pronoella* sp., *Isognomon* (*Mytiloperna*) cf. *bergeroni* Chavan, *M.* (*M.*) *bipartitus* Sowerby, *Myopholas acuticostala* (Sowerby), *Mylilus undulatus* (Sowerby), *P.* (*P.*) *Striatula* (Sowerby), *Solemya woodwardiana* Leckenby, *Anomalodesmatan* sp., *Stenogmus pentagonalis* Cooper, *Ivanoviella cf. sicinbessi* (Quenstedt). *Anisocardia tenera* (Sowerby), *Rollierella minima*, *Grammatodon* (*Grammatodon*) *clathratus* (Leckenby), *Protocardiacon gnata* (Sowerby), *Myopholas douvillei* Lissajous, *Isognomon mytiloides* Lamarck; *Protocardia congnata* (Sowerby), *Modiolus bipartus*, *Modiolus acuticarinatus*, *Astarte multiformis*, *Pronoella* sp., *Sowerby atrianglaris.*	Marine	Callovian–Oxfordian Callovian–Oxfordian Callovian	This study Bai, 1989 Chen et al., 2005
Buqu Fm.	*Actinostreon gregareum* (J. Sowerby), *Pseudolimea duplicate* (J. de C. Sowerby)		Bathonian-Callovian	
	Assemblage *Camptonectes laminatus–Anisocardia beaumonti*: *Camptonectes*, *Gervillella*, *Myopholas*, *Anisocardia*, *Pinna*, *Plagiostoma*, *Rollierilla*, *Brachidontes*, *Gryphea*, *Pholadomya* (*Bucardiomya*) *bucardium* (Moesh), *Ph.* (*B.*) *protei* (Brongniart), *Ph* . (*Pholadomya*) *ovalis* (Sowerby), *Radulopecten vagans.* *Anisocardia* (*Anisocardia*) *beaumonti d'Archiae*, *Camplonecies* (*Camplonecies*) *laminalus* Sowerby, *C.* (*Annulinectes*) *obscurus* (Sowerby), *Prolocardia stricklandi* (Morris et Lycett), *Pholadomya* cf. *carinata* Goldfuss, *Caromyapsis ?siriata* (d' Orbigny), *Gervillella acuta* (Sowerby), *G. ovata* (Sowerby), *Myopholas aculicoslata* Agassiz.	Marine	Middle - Late Bathonian	Chen et al. (2005) Bai (1989)
Upper Quem Co Fm.	Assemblage *Liostrea birmanica–Actinostreon gregareum–Isognomon* (*Mytilopera*) ex gr. *Patchamensis*: *Liostrea birmanica* Reed, *Actinostreon gregareum* (J. Sowerby), *Isognomon* (*Mytiloperna*) ex gr. *patchamensis* Cox, *Isognomon* sp., *Plagiostopma* Sp., *Anisocadia* Sp., *Corbula* sp. nov. *Plicatula* sp., *Burmirhynchia* sp., *B. gutta* Buckman, *Holcothyris subovalis* Buckman, *Myopholas acuticoslata* (Wowerby), Eomiodon angulatus (Morris et Lycett), *Protocardia* (*Prolocardia*) *stricklandi* (Morris et Lycett), *Tancredia extensa* Lycett, *Pleuromya alduini* (Brongniart). Assemblage *Eomiodon angulatus–Isognomon* (*Mytiloperna*) *bathonica*: *Isognomon* (*Mytiloperna*) *muchisoni* (Forbes), *Isognomon* (*Mytiloperna*) *bathonica* (Morris et Lycett), *Eomiodon angulatus* (Morris et Lycett), *Astrata oolitharum*, *Cossmann*, Mytilus (Faclymytilus) sublaeriss, Protocardia trickland.	Brackish-marine Brackish-marine Brackish-marine	Bathonian Bathonian Bathonian	This study Bai, 1989 Chen et al., 2005
Lower Quem Co Fm.	Assemblage *Anisocardia gibbosa–Camptonectes auritus–Pteroperna costatula*: *Anisocardia* (*Anisocardia*) *gibbosa* (Münster), *Ceratomya Bajocian* (d'Orbigny), *Trigonopsis similis* (Münster), *Protocardia truncate*, *Pleuromya subelongata* Agassiz, *Lucina dispecta* Phillips, *Quenstedia oblita* Greppin, *Astarte* (*Astarte*) *elegans* Sowerby, *A.* (*A.*) *minuta* Phillips, *Astarte* (*Leckhamptonia*) *interlineate* (Lycett), *Lycettia lunularis* (Lycett), *Corbicellopsis schmidi* (Greppin), *Thracia lata* Guldfuss, *Kobayashites*; *Camptonecies* (*Camptonecies*) *auritus* (Schlotheim), *Osteomya dilate* (Phillips), *Tancredia truncate* Lycett, *Pteroperna costatula* (Morris et Lycett), *Radulopecten vagans* (Sowerby), *Modiolus* (Modiolus) *anatinus* (Smith). *Astarte* (*Astarte*) *muhibergi* Greppin, *A.* (*A.*) *elegans* Sowerby, *A.* (*A.*) *minima* Phillips, *Thracia* cf. *studeri* Wildborne, *Oxyfoma inaguivaluis* (Sowerby), *Oxyfoma munsteri* Bronn, *Pseudolimea* sp., *Trigonapsis similis* (Münster), *Anisocardia gibbosa* (Münster), *Protocardia lycettina* (Rollier), *P. hepingxiangensis* J. Chen et Lin, *P. truncate* (Goldfuss), *Koboyashites hayamii* Yin, *Corbicellopisis schmidi* (Greppin), *Pleuromya oblita* Greppin, *P.* cf. *marginata* Agassiz. *Camptonectes auritus* (Schlotheim), *Pteroperna costatula* (Deslongchamps), *Anisocardia gibbosa* (Münster), *Astarte elegans* (Sowerby) , *Kobayashites* sp., *Cuneopssis sichuanensis* Gu, Ma et Lan, *Eolamprotu a crerneri*, *Protocardiahe pingxingensis* J . Chen, *Protocardia yanshipingsis* J . Chen, *Astarte minima* (Phillips), *Astarte muhbergi* Greppin. *Corbula? Sinensis* Chen, *Neomiodon yanshipingensis* (Wen), *Eomiodon* Sp., *Protocardia* (*Protocardia*) *Hepingxianensis* J. Chen et Lin, *P.* (*P.*) *Lycettia* (Rollier). Assemblage *Psilunio–Lamprotula* (*Eolamprotula*)*–Cuniopsis*: *Psilunio* sp., *Psilunio sinensis* Gu., *Psilunio henanensis* Gu., *Lamprotula* (*Eolamprotula*) sp. nov., *Lamprotula* (*Eolamprotula*) *subquadrata* Gu., *Cuniopsis johannisboehmi* (Frech)., *Cuniopsis* cf. *sichuanensis* Gu, Ma et Lan., *Unio* sp. nov., gen. et sp. indet., *Corbulomima* cf. *wulanwulaensis* Sha., *Solenaia tanggulaensis* Wen., *Undulatula*? *tanggulaensis* Gu. *Lamprotula* (*Eolamprotula*) *cremeri* (French), *L.*(*E.*) *abrupliscripta* Gu, *Psilunio chaoi* (Grabau), *P. Thailandiscus* (Hayami), *P. ovalis* Ma, *Cuniopsis sichuanensis* Gu, Ma et Lan., *Unio yunnanensis* Ma. *Cuneopsis sichuagensis* Gu, Ma et Lan, *C.* cf. *Nachamensis* (Mansuy), *Unio yunanensis* Ma, *Lamprotula* (*Eolamprolula*) *abrupliscripla* Gu, *Psilunio avalis* Ma, *Margaritifera* sp., *Cuneopsis jahannisboemi* (Frech).	Brackish-marine Brackish-marine Brackish-marine Brackish Fresh water	Bajocian Bajocian Bajocian Bajocian–Bathonian	Yin (1988) Bai (1989) Chen et al. (2005) Yin (1988) This study Yin, 1988 Bai, 1989

limestones, and biolithite limestones (Fig. 3E; see also Fig. 11). Horizontal, ripple, lenticular and herringbone cross-beddings were well developed in the stratigraphy. Mud cracks (at ~320 m; Fig. 4e), thinly bedded alabastrine gypsum (at ~560 m; Fig. 4f) and gypsum crystals (at ~400 m, ~480 m) were observed in the upper part of the unit (Figs. 2 and 3B). The unit has been suggested to have deposited in a tidal flat environment (Tan et al., 2004). Few bivalves are exposed in this unit, most of which are associated with *Anisocardia-Pleuromya uniformis* (J. Sowerby)–*Modiala biparta* of the Callovian or Callovian–Oxfordian Stages in age (Table 1 and Fig. 4i).

The measured Suowa Fm. is mainly composed of 449 m (2004–2453 m in height) of dark gray microcrystalline limestones, grayish marls, light brown sandy limestones and a few dark gray thin beds of biolithite limestones. Marls and limestones frequently appear in the middle–lower part, with thin interbeds of siltstones and sandstones at the bottom, whereas siltstones and mudstones are present in the upper part of the formation. Close to the top of the sampled Suowa Fm. there are several thin layers of striking yellowish bivalve fragmental beds (Fig. 3E). Above the sampled Suowa Fm., another ~80 m of Suowa Fm., which is partially buried, was not sampled. It is mostly grayish calcareous siltstones intercalated occasionally with thin muddy limestones or marlites (see also Fig. 11). The Suowa Fm. contains many bivalves and a variety of other marine brachiopods, Ammonites, Anthozoa, Stromatoporidea, Dinoflagellates and spores of Oxfordian to Kimmeridgian Stages (Table 1). The unit was suggested to be deposited in a shallow marine platform environment (Tan et al., 2004).

The Xueshan Fm. is a ~1000–1300 m thick upward increasing marine-terrigenous to terrestrial sequence of purple to multiple-colored interbeds of mudstones, siltstones and sandstones with bivalves, plant fragments, pollen and spores (Bai, 1989; Hao et al., 1999; Li et al., 2011) formed in a delta environment (Tan et al., 2004; Zhang et al., 2007). However, it is only ~100 m thick in the YSP section, with the main part eroded (Fig. 2). It is also partially buried and thus was not sampled again. It comprises up-coarsening multi-colored interbeds of mudstones, sandstones and siltstones, with bivalves, plant fragments, pollens and spores (refer to Fig. 11).

Paleomagnetic samples were taken at 1–2 m intervals in 2011 and 2013 using portable gasoline-powered drills, which were equipped with a 2.5 cm diameter drill bit and a water-cooling system. Samples were oriented in situ using magnetic compasses. A sun compass was also used for core samples when weather allowed. The differences between the magnetic compass and sun compass were usually within $\pm 2°$. A total of 1533 paleomagnetic samples were collected. In addition, for cross-calibration purposes, each sample was further cut into three parallel sets of specimens in the laboratory.

4. Stratigraphic chronology

4.1. Biostratigraphy

A number of marine to fresh water mollusks and floras were found in the YSP section, in which bivalves and Brachiopoda are most abundant. The combination of our findings with former studies of the fossils in the YSP section reveals five biostratigraphic zones, with ages ranging from the Bajocian to the Kimmeridgian, indicating two cycles of transgression–regression (from Quem Co Fm. to Xiali Fm. and from Suowa Fm. to Xueshan Fm.) (Table 1 and Fig. 2; see also Fig. 8).

In the lower Quem Co Fm., we found abundant bivalves, which can be classified as two types of assemblage, fresh water and brackish to marine water. The fresh water bivalves mostly occur in the purple-red sandstones to siltstones and are characterized by the assemblage of *Psilunio–Lamprotula* (Eolamprotula)–*Cuniopsis* of the typical Middle Jurassic (Yin, 1988) (Table 1 and Fig. 4g). They were further verified by very low values of oxygen and the carbon isotopes of the bivalve crusts, in contrast to the high values of marine bivalves (Yin, 1989). The brackish-marine bivalves mostly occur in the greenish–yellowish siltstones and mudstones and are dominated by the assemblage *Anisocardia gibbosa–Camptonectes auritus–Pteroperna costatula*, in which many are typical and common components of the Bajocian in age (Table 1). For example, the widely distributed Asia-specific *Kobayashites* sp. occurred only in the Bajocian and *C. auritus* (Schlotheim) predominated overwhelmingly in the European Bajocian stratigraphy and was almost absent from the Bathonian (Yin et al., 1989).

In the upper Quem Co Fm., two assemblages, *Liostrea birmanica–Actinostreon gregareum–Isognomon* (*Mytilopera*) ex gr. *Patchamensis* and *Eomiodon angulatus–Isognomon* (*Mytiloperna*) *bathonica* are dominant, suggesting an age in the Bathonian. The occurrence of typical Bathonian bivalves *Isognomon, Eomiodon, Prolocardia* and *Astarte* characterizes the stratigraphy and suggests a variety of salt degrees from brackish to near marine (Table 1) (Bai, 1989; Yin, 1989; Chen et al., 2005).

In the Buqu Fm., several bivalves and Brachiopoda were found in the limestones and are represented by assemblage *Camptonectes laminatus–Anisocardia beaumonti*, which are similar to those in the upper Quemo Co Fm., but obviously increasing components prevailed in the late Bathonian and Callovian (Bai, 1989; Chen et al., 2005) (Table 1).

The Xiali Fm. contains some bivalves indicated by the assemblage *Anisocardia* ex gr. *Islipensis/tenera–Modiala biparta–Isognomon* cf. *Bergeroni* that lived in a broad time range from the Bathonian to the Oxfordian, such as *Myopholas douvillei* Lissajous, *Isognomon mytiloi des* Lamarck (Callovian), *Grammatodon* (*Grammatodon*) *clathratus* (Leckenby), *Protocardia cognata* (Sowerby), *Anisocardia tenera* (Sowerby), and *Modiolus bipartus*, *Modiolus acuticarinatus*, *Astarte multiformis*, and *Pronoella* sp., *Sowerbya trainglaris*. This indicates that it is unlikely to use them to constrain the boundary between the Bathonian and Callovian. Taking into account the dominant typical Oxfordian bivalves and Brachiopoda observed in the overlying Suowa Fm. and the many Oxfordian-lasting bivalves that occurred in the Xiali Fm., the Xiali Fm. is best constrained in the Callovian–Oxfordian (Table 1).

Bivalves flourished again in the Suowa Fm. Abundant bivalves represented by assemblage *Gervillella aviculoides–Radulopecten fibrosus* dominate the lower and middle Suowa Fm., in which although many also belong to the Callovian–Oxfordian Stages, definite Oxfordian bivalves increase considerably upwards (Table 1). For example, *Chlamy* (*Radulopecten*) *midas* (d'Orbigny) and *Mytilus angulatus* (Young et Bird), *Gervillella aviculoides* (Sowerby) are typical predominant components of the Oxfordian in the Oxfordian stratigraphy of the United Kingdom (Arkell, 1933). Close to the uppermost Suowa Fm. in the YSP section, there are several distinct thin layers of accumulated fragments of *Entolium corneolum* (Young et Bird), *Myopholas multicostata* Agassiz and *Myopholas percostata* Douville (Fig. 4j; see also Fig. 11) that appeared in the Oxfordian but prevailed in the Kimmeridgian, suggesting the top of the Suowa Fm. is in or at least close to the Kimmeridgian (Table 1).

In the lower Xueshan Fm. in the YSP section, the bivalves *Myopholas multicostata* (Agassiz), *Pleuromya* sp. and *Sinonaia* sp. were found, clearly indicating a Kimmeridgian age (Bai, 1989). The middle and upper parts of the Xueshan Fm. eroded away in the YSP section but appear in the Quem Co area, just to the NW of the YSP section, and also in the YSP syncline (Fig. 3A). Some bivalves, including *Peregrinoconcha yunanensis* and *Peregrinoconcha perlonga* of the Tithonian, with a few recognized as the Berriasian, were found in the middle Xueshan Fm. (Bai, 1989) (Table 1).

A variety of other types of fossils, such as marine brachiopods, ammonite, anthozoa, stromatoporidea, dinoflagellates and spores, were also found in the YSP section or other nearby sites, which provides cross-constraints and confirmed the bivalve biostratigraphy above (e.g., Sun, 1982; Yin, 1988; Bai, 1989; Yin, 1989; Zhao et al., 2001; Li and Batten, 2004; Chen et al., 2005; Yi et al., 2005; Cheng and He, 2006; Li et al., 2011).

For example, brachiopods thrived in the YSP section. The earliest fossil brachiopods found in the YSP section lie in the upper Quem Co Fm. and are characterized by an assemblage of *Burmirhynchia gutta* Buckman–*Holcothyris subovalis* Buckman, comprising mainly *Burmirhynchia gutta* Buckman, *Burmirhynchia* sp., *Holcothyris subovalis* Buckman, *Holcothyris rostrata* Buckman, *Holcothyris trigonaris* Buckman, and *Lingula* sp. (Bai, 1989; Chen et al., 2005) (c.f. Fig. 11). This is the same type of Bathonian assemblage in China, indicating a non-normal (restricted) sedimentary environment (Sun, 1982). The normal marine assemblage *Kallirhynchia-yaxleyensis* (Davidson)–*Pseudotubithyris globate* (Sowerby) of the Bathonian was found in the Buqu Fm. (Bai, 1989; Chen et al., 2005). The brachiopods *Stenogmus pentagonalis* Cooper and *Ivanoviella* cf. *sleinbessi* (Quenstedt) found in the Xiali Fm. suggest a wide age of the Callovian–Oxfordian (Bai, 1989).

Dinoflagellates also thrived in the YSP section. Many definite marine dinoflagellate cysts, such as *Amphorula delicata*, *Amphorula metaelliptica*, *Batioladinium* sp., *Glossodinium dimorphum*, *Gochteodinia* sp., *Gonyaulacysta* sp. cf. *G. dualis*, *Muderongia* sp. and *Scriniodinium crystallinum*, were studied in detail in the uppermost Suowa Fm. and lower Xueshan Fm., suggesting ages of the Oxfordian–early Kimmeridgian and even to the Berriasian in the Early Cretaceous (Cheng and He, 2006; Li et al., 2011).

4.2. Magnetostratigraphy

Combined with our previous work on the upper YSP section (Song et al., 2016), paleomagnetic measurements were applied to the whole YSP section. Additionally, related rock magnetic measurements of representative samples were performed to understand the types and behaviors of the magnetic minerals that carry remanences for paleomagnetic directions and reliability.

The paleomagnetic and rock magnetic measurements were performed in the paleomagnetic laboratories of Lanzhou University, China University of Geosciences (Beijing) and the Institute of the Tibetan Plateau Research, Chinese Academy of Sciences (CAS) in magnetically shielded rooms (<250 nT). For rock magnetic characterization, low-field temperature-dependent susceptibility (κ–T) was analyzed on a Kappabridge MFK-1 magnetic susceptibility meter coupled to a CS-4 temperature control apparatus in an inert argon atmosphere at the ITPCAS. Both alternating field demagnetization (AFD) and thermal demagnetization (TD) were performed for characteristic remanent magnetization (ChRM) isolation, and the remanent magnetizations were measured by 2G Enterprises magnetometers.

Two magnetization categories were observed based on the κ–T curves: (1) hematite is the principal magnetic carrier, as indicated by a clear Curie temperature at ca. 680 °C but a less clear susceptibility loss near 580 °C (Fig. 5A, D); and (2) magnetite and hematite are the primary magnetic carriers, as suggested by the sharp loss of susceptibility near 580 °C and a further loss at ~650–670 °C (Fig. 5B, C).

Thermal demagnetization analyses revealed two magnetic components for most of the specimens: a low-temperature component (LTC) and a high-temperature component (HTC). The LTC was usually removed at temperatures less than 200–300 °C, with the mean consistent with the present day geomagnetic field, indicating the presence of a soft viscous overprint. The HTC reveals a stable demagnetization path from 400 °C to 580 °C and/or 690 °C for most specimens, pointing to the origin in the orthogonal plot (Fig. 6A-C and F), which indicated the magnetic carriers of magnetite or both magnetite and hematite, respectively. The AFD showed that the lower field component was clearly removed at ~10 mT, indicating clearance of viscous remnant magnetizations (VRM), whereas the higher field component, which decayed toward the origin by ~40–50 mT, showed a clear characteristic remnant magnetization direction carried mostly by soft magnetic minerals (Fig. 6D-E). Together with the κ–t curves, both the thermal and AF demagnetizations indicated that the main magnetization carriers are mostly magnetite, and some are mixtures of magnetite and hematite in the YSP section.

Characteristic remanent magnetizations (ChRMs) were obtained by principal component analysis. Samples with random directions, unblocking temperature less than 350 °C, or ChRM directions with maximum angular deviation >15° were abandoned for further analyses. A total of 864 out of 1533 samples yielded reliable results and were used for magnetostratigraphic analyses. In addition, given the existence of transitional polarity directions, a criterion of virtual geomagnetic pole (VGP) latitude of 45° was used for the mean direction calculation and field tests. We chose a total of 591 specimens having robust polarity directions (i.e., VGP-Lat >45°) for the reversal test, yielding means of $D_s = 354.4°$, $I_s = 40.9°$, $\alpha_{95} = 2.1$, $k = 12$, and $n = 377$ for the normal polarity directions and $D_s = 177.2°$, $I_s = -42°$, $\alpha_{95} = 3.9$, $k = 9$, and $n = 159$ for the reversed polarity directions (Fig. 7A), demonstrating a positive Tauxe (1998) reversal test with McFadden and McElhinny (1990) classification of A ($\gamma c = 4.12°$) and a positive McElhinny (1964) fold test at the 95% confidence level, with the best grouping at 92.4% unfolding (Fig. 7C).

Magnetostratigraphy was defined by plotting directions as an angle from the mean virtual geomagnetic pole (VGP) of the entire dataset. Each polarity was defined by at least two subsequent levels with the same polarity. In total, 66 normal and 65 reversal polarities were identified throughout the entire section and were labeled as N1–N66 and R1–R65. The reliability test of the observed polarity yielded $J = -0.2643$, which fell within the recommended range of 0 to −0.5 for a robust magnetostratigraphic data set (Tauxe and Gallet, 1991; Supplement material A), indicating that our samplings have recovered robust magnetostratigraphic patterns.

In accordance with the biostratigraphic constraints above (Table 1), the obtained magnetic polarity zones of the YSP section were best correlated with chrons M28–M45 of the Geomagnetic Polarity Time Scale (GPTS) (Gradstein et al., 2012) (Fig. 8). Specifically, normal zones N01–N08 were most likely correlated to the interval of M28 to M30, strikings N12–N14 to long M32–M33 more or less equally occurred in the four normal zones N15–N18, followed by a long reversal zone R18 to M33A–C, which more or less equally occurred in the eight normal zones N19–N26 to M34–M36A–C. The very striking long normal zones N27–N34 to M37–upper M39, more or less alternately occurred in N33–N49, with reversals to lower M39–uppermost M42, which densely occurred in short normal zones N54–N65 to M43–M44 and the bottom

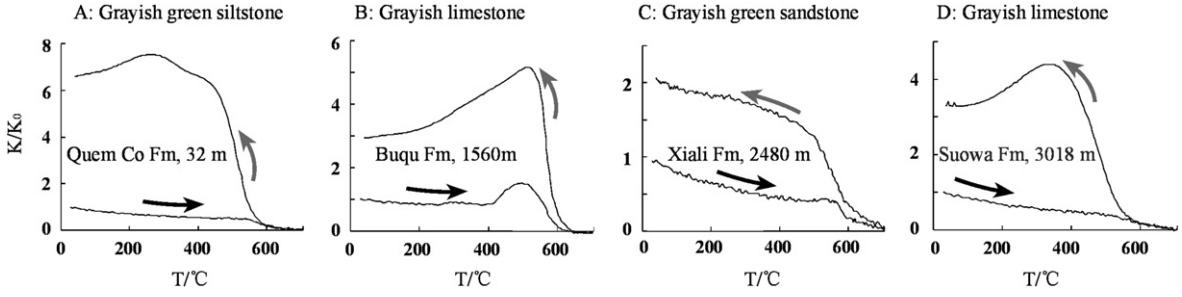


Fig. 5. Representative κ–T curves for the YSP section. Black and gray arrows represent the heating and cooling curves, respectively.

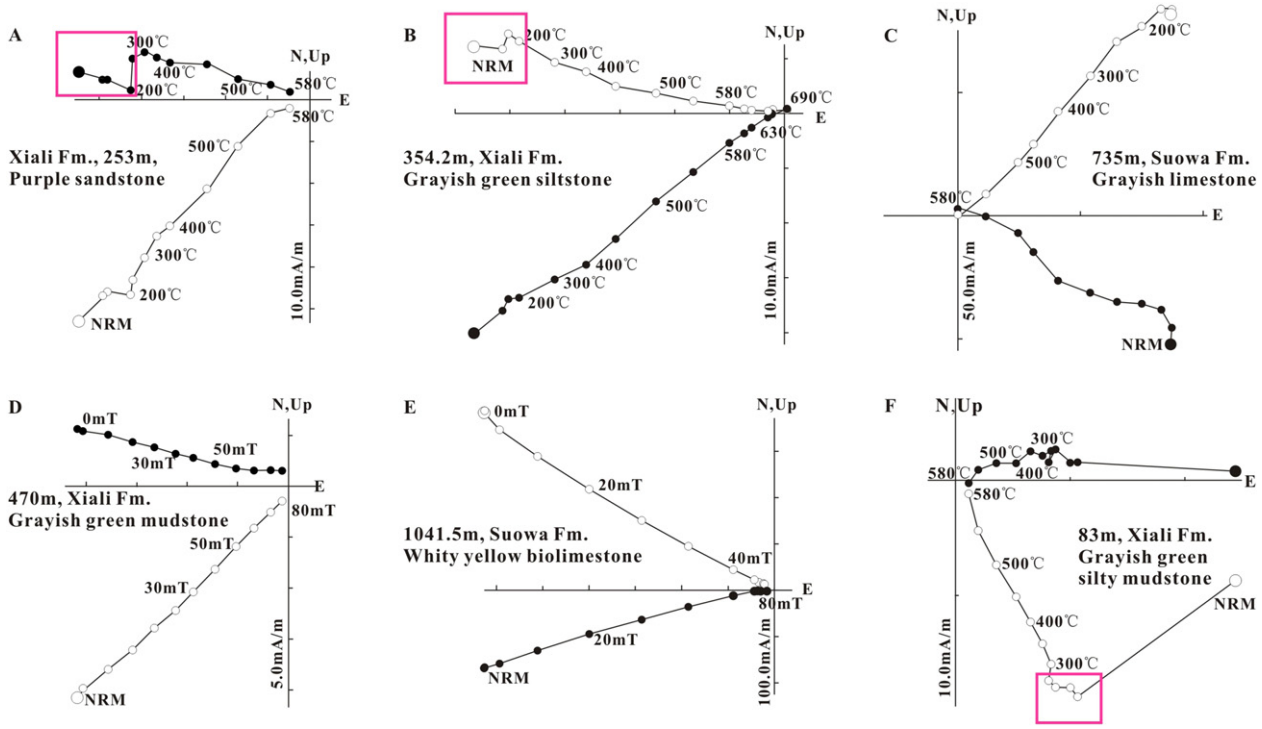


Fig. 6. Orthogonal projection diagrams of representative TD and AFD in the YSP section. Directions are plotted in situ. Open (closed) symbols represent vertical (horizontal) projections. NRM is the natural remanent magnetization.

marked long normal N66 well to the striking long polarity zone M45 (Fig. 8).

This correlation yielded magnetostratigraphic ages of >171.2–<157.5 Ma (Late Aalenian to Early Oxfordian of the Middle–Late Jurassic) for the YSP section and constrained the Quem Co, Buqu, Xiali, Suowa and Xueshan Fms., respectively, to approximately >171.2–165.5 Ma (late Aalenian to early Callovian), 165.5–163.3 Ma (middle to late Callovian), 163.3–160.1 Ma (early Oxfordian), and 160.1–<157.5 Ma (late Oxfordian), with the top unmeasured YSP section lasting to the early stage of the Kimmeridgian (157.3–152.1 Ma) (Fig. 8). This magnetostratigraphic age determination is further supported by the high correlation coefficient (R) and constant sedimentary accumulation rates (SAR) throughout the whole section (Supplement material B). However, we also note that the correlations of N50–N65 to M42–M44 are not a perfect one-to-one match, possibly due to the larger sampling intervals or some incomplete isolation of ChRMs.

5. Working model

Based on the basic principles of physical geography, for an ideal ocean (no continent) earth, the combined roles of the heating zonation of the earth due to the earth's tilting and varying distances between the earth surface and the sun and the Coriolis force lead to the global circulation system, i.e., the Hadley cell in the equator to the subtropic and the Fishier cell from the subtropic to the polar. In the Hadley cell, the equator strongest insolation and heating drive the warmest moistures to the upper troposphere and move northward to the subtropic zone and then sink, resulting in low pressures and humid-hot climate in the equator and tropic zones, high pressures and dry-hot climate in the subtropic zones, the northeast trade winds in the subtropic-tropic zones and the easterly winds near the equator. The intertropical convergence zone (ITCZ), with the lowest pressure (or strongest convectional rise and thus heaviest precipitation), is located near the equator without clear fluctuation (Fig. 9). In the Fishier cell, further northward moving airs mostly turn to the westerly winds in the midlatitudes (temperate) and partly flow to and sink in the polar region as high pressure cold air, causing easterly and NEE winds in the lower levels. These winds are unstable and frequently interrupt the westerlies, causing strong cyclonic-frontal rains (Fig. 9).

However, the existence of continents interferes with this ideal global circulation and climate system, breaking the homogeneous zones as belted air masses or replacing them as regional monsoon circulation, depending on the sizes and positions of the continents (Fig. 10). The most significant changes are in the tropic–subtropic and midlatitude zones, where the subtropic high and part of the westerlies are broken on large continents, such as Eurasia, in summer and are intensified in winter. This causes strong air mass interactions between the marine subtropic highs and westerlies and the continents and between the oceans and the continents (Fig. 10).

For the eastern coast of a large continent in the tropic to subtropic zones, in summer, the great heat contrast between the large continent and the ocean gives rise to hot, low pressure over the continent and the warm, relatively high pressure over the ocean. This stimulates weak subtropic monsoon (STM) circulation (thus raining) in the southeastern (or northwestern) periphery of the subtropic high in the northern (or southern) hemisphere and its nearby eastern continent, i.e., southeast monsoon (SEM) in the northern hemisphere or northwest monsoon (NWM) in the southern hemisphere (Fig. 10a, b). The modeled East Asian monsoon (EAM) climate in the subtropic without the Himalaya–Tibetan Plateau (e.g., Kutzbach et al., 1989; Liu and Yin, 2002; Wu et al., 2009) is a good example of such a subtropic monsoon. Similarly, if the large continent in the northern hemisphere extends more southwards into the tropic and even to the equator, the considerably enhanced sensible heating due to the extended land in the tropic and the equator absorbing much more solar energy will induce a much stronger tropic monsoon (TM) (Dirmeyer, 1998; Liang et al., 2005). If the heating low pressure is low enough, it may attract the southeast trade winds and the easterlies in the southern hemisphere northward into the northern hemisphere to form a stronger trans-equator tropic monsoon (TEM), which will be self-enhanced by the great latent heating transported from the southern hemisphere and lower tropic ocean when raining (Liang et al., 2005). Thus, the TEM or

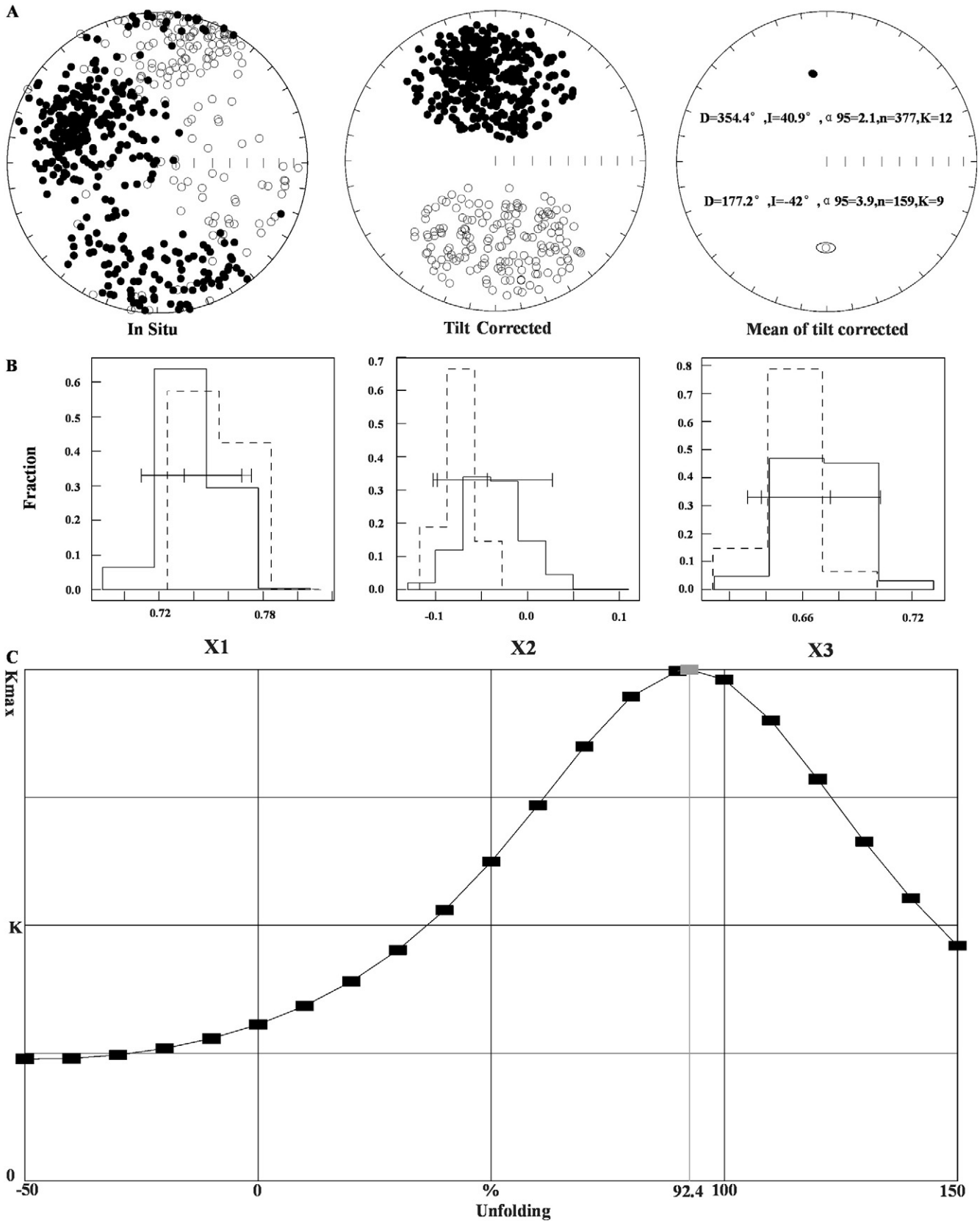


Fig. 7. (A) Equal-area projection of the ChRM directions and mean directions (with 95% confidence) for the YSP section determined by the bootstrap method (Tauxe, 1998); downward (upward) directions are shown as filled (open) circles; (B) the histograms of the Cartesian coordinates of the bootstrap reversal test. The intervals containing 95% of each set of components are shown (dashed and solid lines). The confidence bounds from the two data sets overlap in all three components, which suggests that the means of the reversed and normal modes cannot be distinguished at the 95% level of confidence, indicating a positive bootstrap reversal test (Tauxe, 1998); (C) The McElhinny (1964) fold test for the obtained ChRMs indicates that at the 95% confidence level, Ks/Kg = 4.03 > critical value 1.13 (F (2 * (n2 − 1),(n1 − 1)), with a maximum grouping at 92.4% unfolding (Fig. 7C), yielding a positive fold test.

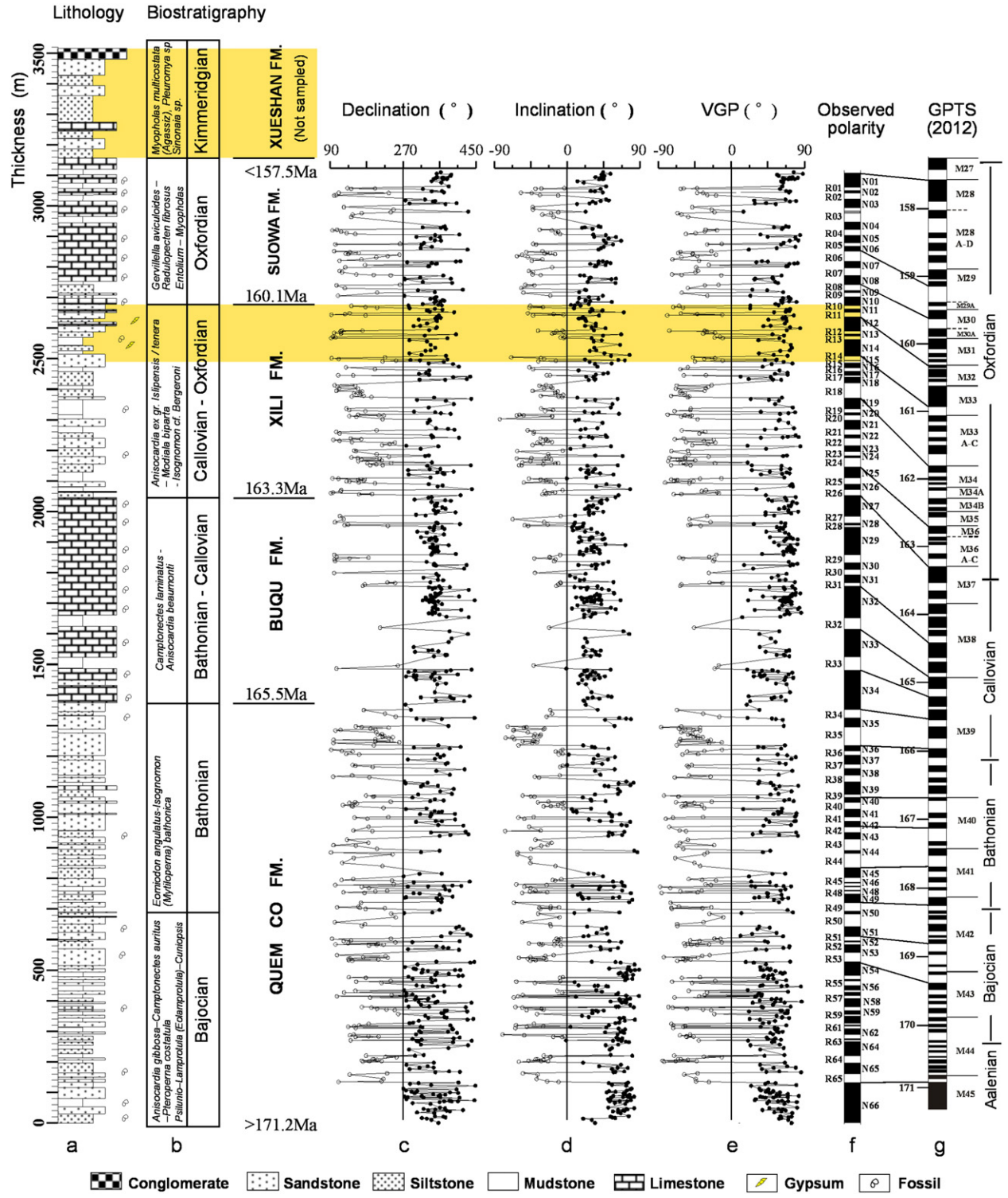


Fig. 8. Magnetostratigraphy vs. lithostratigraphy of the YSP section.

tropic monsoon (TM) is much stronger than the STM. If the continent further extends southwards across the equator into the southern hemisphere, the above processes and TEM will be enhanced and stabilized (Fig. 10c, d). However, it will weaken the moist southeastern trade winds and thus dry the hinterland of the continent in the southern hemisphere (Fig. 10c, d). Numeric modeling of the Indian monsoon (IM) and EAM without the India, Indochina and African continents (both without the Himalaya-Tibetan Plateau) shows no trans-equator current IM and EAM in the tropic but weak EAM in the subtropic, confirming the theoretical analysis above (Liang et al., 2005).

In winter, in contrast to a warm ocean, the center of the large continent in the subtropic becomes cold, resulting in high pressures and strong outflowing cold-dry winds (winter monsoon) on the continent (Fig. 10).

For the western part of the large continent in the subtropic, in summer, the continental low pressure attracts cold-dry air masses from high

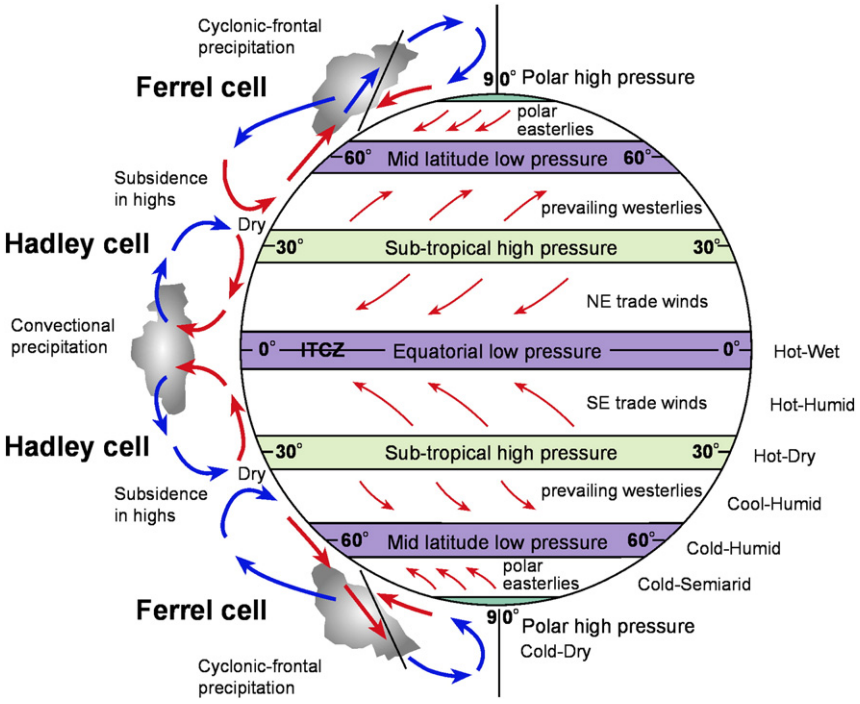


Fig. 9. Schematic diagram showing the global circulation pattern of an ideal ocean earth.

latitudes (Wu et al., 2009) together with the subtropic high, leading to a warm/hot-dry climate. In winter, the continent cold, high pressure also leads to a dry climate. This year-round dry climate gives rise to desert formation in the center and western parts of the continent (Wu et al., 2009) (Fig. 10).

However, for the west part of the large continent in the equator and tropic zones, a TEM originating from the southern subtropic high is formed due to seasonal fluctuation of ITCZ caused by the sun (Ruddiman, 2008) and by the attraction of the low pressure in the tropic continent, bringing heavy rains in summer (Fig. 10d).

Theoretical analyses and numeric modeling demonstrated that buildup of large E–W mountains, such as the Himalayas, or a plateau, such as the Tibetan Plateau, in the tropic and subtropic could considerably enhance the heating effect on the continent and thus further intensify the summer and winter monsoons (both TM and STM) because high mountains or plateaus can absorb more sensible heating and latent heating from rain on the southern slope than low land of the same latitude in the summer and become colder, thus having higher pressure than low land of the same latitude in winter (e.g., Ye and Gao, 1979; Kutzbach et al., 1989; Ruddiman and Kutzbach, 1989; Liu and Yin, 2002; Wu et al., 2009; Boos and Kuang, 2010) (Fig. 10b). Meanwhile, mechanical hindering of moist air in the continental center and possible bending of the westerlies (thus forming anticyclone-like sinking airs) further dry the central and western parts of the continent (Kutzbach et al., 1989; Ruddiman and Kutzbach, 1989).

For the midlatitude zone, when the westerlies penetrate into a large continent, such as Eurasia, the moist air mass becomes progressively drier due to on-the-way precipitation, but it would not be as dry if there were no mountains on the western continent coast or plateaus on the continent, such as Tibetan Plateau (Manabe and Broccoli, 1990; Dirmeyer, 1998) (Fig. 10). If south–north stretching mountains occur on the west coast of the continent in the midlatitudes, all-year raining will occur on the windward (western) slope and foehn wind and dry climate will occur on the leeward (eastern) slope and its adjacent area. The higher the mountains are, the more enhanced and extended the foehn wind and dry climate on the leeside (Fig. 10).

Between the midlatitude and subtropic zones, south–north migration of the westerlies and subtropic high due to seasonal change will form the so-called Mediterranean climate (Ruddiman, 2008), with precipitation in winter due to the westerlies and dryness in summer due to the subtropic high (Fig. 10).

If the continent at the equator and lower tropic zones is small, such as the Caribbean Islands and Central America, or medium, such as Australia and South America, the heat contrast between the continent and ocean is not sufficient to stimulate a stable monsoon circulation and the easterlies and trade winds play dominant roles. They blow and bring heavy rains either through the whole continent or only to the east coast of the continent, forming a tropic rain forest climate (Fig. 10a, b). The trade winds can further act on the east coast of the subtropic continent, causing a subtropic monsoon-like humid climate (Fig. 10a, b).

However, this is exceptional for the equatorial and tropic continent, where a TEM occurs. The TEM brings moist air not directly westwards into the continent interior but northwards into the northern hemisphere, where the air flow turns to the southwest direction off the equatorial and tropic continent, leading to drying of the continental interior (Fig. 10c, d). The present tropic savanna climate in the interior of equatorial and tropic Africa is a good example, where the trans-equator Indian monsoon carries moist air parallel to the eastern Africa shoreline northwards into the northern Indian Ocean with southwest flow (Fig. 5).

Regardless of whether a stable monsoon or monsoon-like climate occurs in the eastern continent, the enhanced condensation and latent heating in the east coast and reduced heating in the nearby ocean form a local cycle that intensifies the monsoon (Wu and Liu, 2003; Wu et al., 2009).

The ocean currents are driven by air circulation, but in turn can enhance or weaken the air circulation and climate. The major currents are the easterly warm current in the equator and tropic zones driven by the trade winds and easterlies and the westerly cold current driven by the westerlies in the mid and high latitudes (Fig. 10). When these currents meet on the continent, the warm easterly current turns, flowing from low to high latitudes (polewards) along the east coast of the continent, enhancing ocean evaporation and moist air (climates), while the cold current flows from high to low latitudes (equatorwards) along the west coast of the continent, suppressing ocean evaporation and intensifying the dry climate on the west coast (Fig. 10).

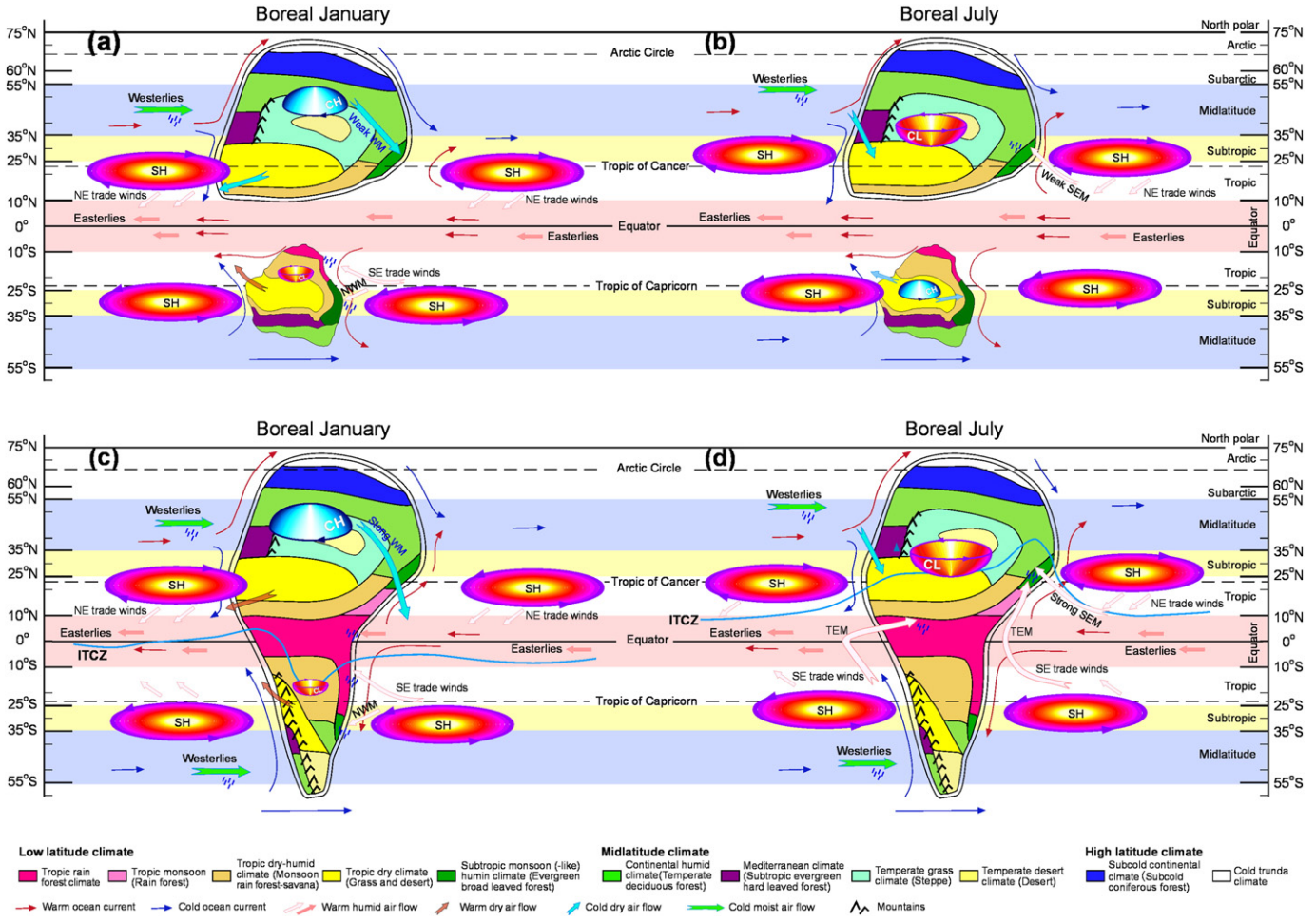


Fig. 10. Schematic diagram showing the interaction of the ideal global circulation pattern in Fig. 9 with hypothetical ideal continents and mountains based on the principles of physical geography and conceptual and numerical modeling, giving rise to interacting global circulation models that determine various types of climates. Note that the northern hemisphere summer is just the reverse of the southern hemisphere winter. TEM on the west coast is limited to only the tropic zone (tropic monsoon). Only strong heating on a large continent can attract the TEM tropic monsoon further northwards to the subtropic and midlatitude zones (subtropic and temperate monsoons). ITCZ, Intertropic convergence zone; SH, Subtropic High; CH, Continental High; CL, Continental Low; TEM, Trans-equator monsoon; SEM, Southeast subtropic monsoon; NWM, Northwest subtropic monsoon or monsoon-like subtropic humid climate.

The interactions of modern continents and mountains with physiographic heating zones and their determined air and ocean circulations and climates provide a way to test the working models above (Fig. 11). The trade winds and easterlies bring heavy rains throughout small and medium size continents in the tropic and equator zones, rain occurs only on the east coast of relatively large continents in the poleward part of the tropics, such as northeast Australia and central east South America, forming a tropical rain forest climate (Fig. 11). Northeastern or monsoon-like moist winds emitted from the southwestern rim of the subtropic high in the southern hemisphere bring rains to the eastern and part of the southeastern coast of the continent, causing subtropic monsoon-like or humid climate, for example, the eastern and part of the southeastern coasts of Africa, Australia and South America (Fig. 11). The southeast coast of North America in the subtropics mimics those in the southern hemisphere. The exception is for the southeastern and eastern Asia coasts, where strong trans-equator monsoons (Indian and East Asian monsoons) occur, covering south and Southeast Asia and reaching the northeastern Asian coast to form a temperate monsoon climate (Fig. 11). The trans-equator Indian monsoon has further caused drying of the eastern Africa coast in the tropic and equator zones, where there should have been a rain forest climate based on the trade winds and easterlies, where no trans-equator monsoon would occur (Fig. 11). The other trans-equator tropic monsoon is formed around the equator and tropic western Africa coast, i.e., the North African monsoon (NAM) due to the large amplitude north–south fluctuations of the ITCZ with season, forming a tropic monsoon rain forest climate (Ruddiman, 2008) (Fig. 11). For the other parts of the subtropic-tropic zones, tropical desert climates predominate (Fig. 11).

The midlatitude zone in the northern hemisphere is characterized by a temperate cool-humid marine climate on the western coasts of the continents and western slopes of mountains and by temperate continental cool/cold-subhumid to desert climates on the continent and its interior. However, high mountains, such as the Rocky Mountains, and high plateaus, such as the Tibetan Plateau, play a critical role in greatly intensifying dryness in the continental interior because their high topography blocks moist airs from entering the continental interior and bends the westerlies and induces anticyclonic sinking cold-dry air on the southeastern (leeward) side (Kutzbach et al., 1989; Ruddiman and Kutzbach, 1989). To the north and south of the westerly zone, cold-dry polar climates and Mediterranean climates (hot-dry summer and cold-humid winter) dominate, except for monsoon-impacted regions (Fig. 11). Surface ocean currents are driven mostly by trade winds and westerlies. When they meet continents, they transport heat from the equator and tropics to the midlatitudes and bring cold water from the high-mid latitudes into the tropics, thus enhancing the moist and dry air (climates) on the continental eastern and western coasts (Fig. 11).

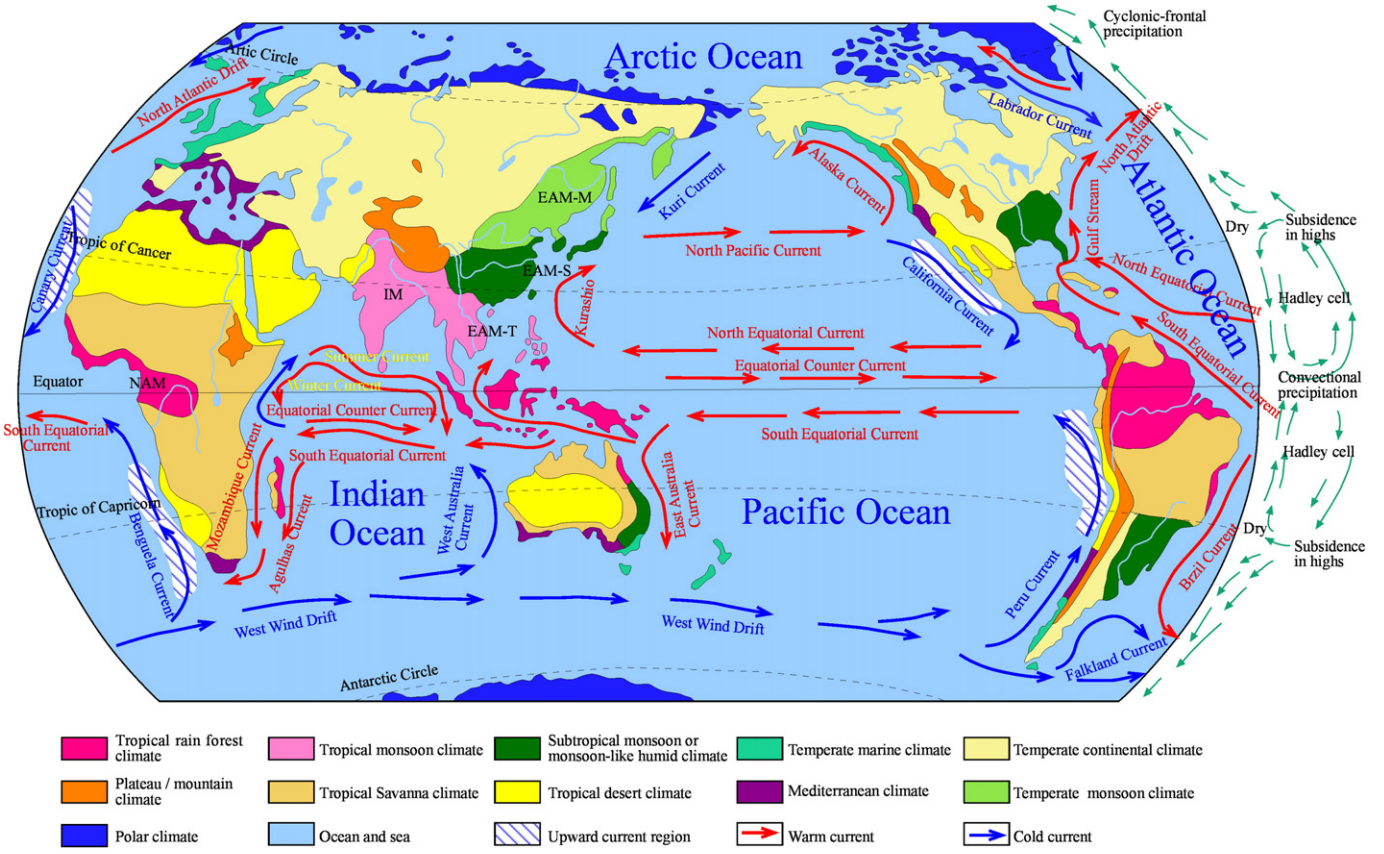


Fig. 11. Verification of the model predicted circulations and climates in the world climate distribution map. IM, Indian monsoon; EAM-T, East Asian tropic monsoon; EAS-S, East Asian subtropic monsoon; EAS-M, East Asian midlatitude (temperate) monsoon; NAM, North Africa tropic monsoon.

Our proposed models can interpret modern climates very well, verifying their reliability.

6. Climatic record and change in the Qiangtang Basin

We combined the lithologic climatic records of the QB from our field investigations and the literature (e.g., BGMX, 1982; Hao et al., 1999; He et al., 2002; Fang and Li, 2005; Wang et al., 2007b; He et al., 2011) (Fig. 2). The Late Paleozoic climate warmed from the early glacial cold-dry climate to a hot-humid climate, as indicated clearly by the occurrence of marine tills and cold fauna in the Late Carboniferous–Early Permian and warm fauna and coals in the Late Permian (Fig. 2). The climate in the Triassic succeeded the hot-humid Late Permian, as demonstrated by the continuous occurrence of coals and red beds, except for the Middle Triassic, when gypsum layers occurred in the bay and lagoon along the NQB marginal areas and in the Tumengela depression (e.g., Wang et al., 2009; He et al., 2011), indicating an interposed dry climate (Fig. 2). Coals continued to occur in the Early and Middle Jurassic, suggesting a persistent hot-humid climate at that time (Fig. 2). Approaching the earliest Late Jurassic (early Oxfordian) approximately 161 Ma, a thin gypsum layer appeared in the red beds of the upper Xiali Fm. in the Yanshiping section (Figs. 2–4). Moreover, the red beds and intercalated gypsum layers predominated the Late Jurassic and Cretaceous and lasted into the early Cenozoic (Fig. 2). Thus, the occurrence of gypsum and red beds in the Late Jurassic indicates a shift of the late (J_3-E) persistent dry-hot climate to the former dominant humid-hot climate since the Late Permian (Fig. 2).

7. Evolution of megamonsoon, mechanisms and discussion

Using the working models above (Fig. 10), we interpreted the QB late Paleozoic to Mesozoic climatic changes as responses to global and QB continental drifts and collisions and megamonsoon onset, evolution and retreat/end (Fig. 2).

Fig. 12 summarizes the state-of-the-art paleomagnetic data of the QB from Yan et al. (2016). Although the Paleozoic-Mesozoic paleomagnetic data are relatively scarce and some are poor, these data, together with the geological and paleontological evidence, show that the QB was part of the northeastern margin of the Gangwana at approximately 20°S in the subtropic high during most of the Paleozoic and broke away in the Early Permian (Li et al., 2009; Zhu et al., 2013). It moved very fast northwards from the southern to northern hemispheres, reaching the tropic in the northern hemisphere in the Late Permian and Early Triassic and the subtropic in the Middle Triassic. Since then, the QB has remained in the subtropic zone (Fig. 12).

The birth of the Pangea continent in the Late Permian from the conjunction of the Laurasian and Gondwana continents by the Hercynian Movement resulted in the greatest heat contrast between the continent and ocean and thus initiated the strongest monsoon in the earth's history, the megamonsoon (Kutzbach and Gallimore, 1989; Weissert and Mohr, 1996; Rais et al., 2007). According to the working model above, in summer, the huge Laurasian continent would have created the strongest heating and thus the lowest air pressure in the continental interior in the tropic-subtropic zones, while the huge Gondwanaland would have formed the highest pressure in its interior in the subtropic-midlatitudes due to the very cold temperature in the interior during the southern winter. The subtropic high would have been broken over the continent and preserved over the Paleo-Pacific Ocean. This configuration of the air masses and pressures would have led the Laurasian continental lowest pressure center (CL) to exert great attraction on the flows emitted from the Gondwana continental highest pressure center (CH) that flew northward and across the equator and turned to the northeast to the southeast side of the CL to form the trans-equator megamonsoon to bring a large amount of warm moist and precipitation

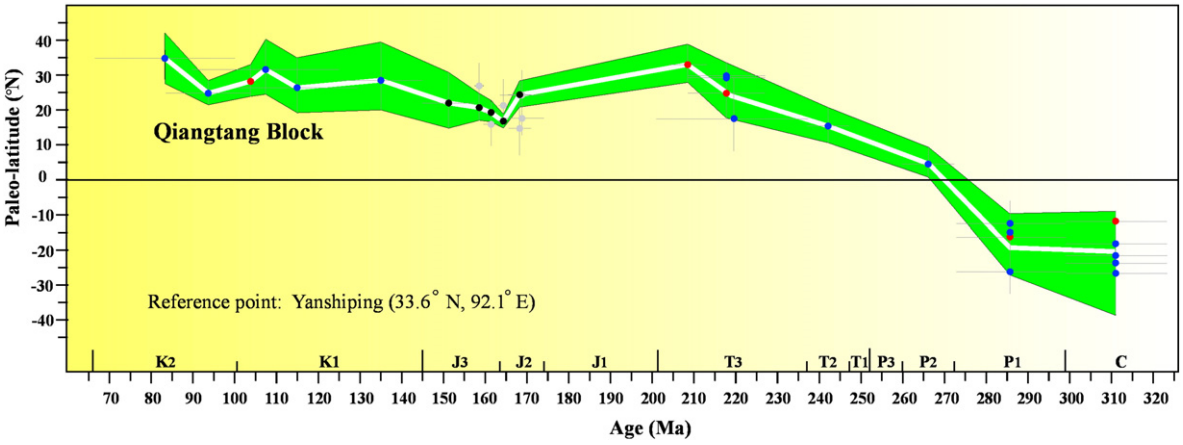


Fig. 12. Paleolatitude evolution of the Qiangtang Block since the late Paleozoic. (Modified from Yan et al. (2016) with newly updated data).

from the south and north Paleo-Tethys warm ocean to the coast areas of the Paleo-Tethys and form a tropic rain forest climate, leaving a very dry Pangea interior (Fig. 13). The north subtropic high over the Paleo-Pacific Ocean could have extended its range to the east part of Paleo-Asia, causing dry climate, but the trade winds could have brought some moisture and precipitation to the east coast of Paleo-Asia to the south of the high (Fig. 12). In winter, this pattern of air pressure and climate was reversed, with precipitation along the east and northeast coasts of Gondwanaland (Fig. 13). Geological evidence (Permian coals around the Paleo-Tethys coast and widespread deserts and salts in the Pangean interior) (e.g., Boucot et al., 2009 and references there in; BGMX, 1982; Hao et al., 1999) and numeric modeling (Kutzbach et al., 1989) confirmed this theoretical analysis. Our records provide new evidence. Drifting of the QB toward the equator and north tropic zones (megamonsoon impacting regions) in the Late Permian–Early Triassic and later into the subtropic zone in the Late Triassic led to the megamonsoon and easterly winds and brought heavy precipitation through the small QB, likely forming a tropic rain forest climate and coals in the Late Permian–Early Triassic. In contrast, the subtropic high caused a dry climate, forming gypsum and red bed deposition in the QB in the Late Triassic (Figs. 2, 12, 13). The occurrence of Late Permian–Early Triassic Cathaysia flora and coals (Hao et al., 1999) and the wide distribution of Middle Triassic gypsum and salts in other blocks near the QB (such as the Qadam-Simao and Yangtze Blocks) (BGMX,1982; Hao et al., 1999; Boucot et al., 2009) provide further support (Fig. 13c).

From the Late Triassic to the Middle Jurassic, the QB returned to a hot-humid climate (Fig. 2). Because the position of the QB was not subjected to significant changes and remained in the subtropic high, we proposed that the enhanced and eastward expansion of the megamonsoon could bring heavy precipitation to the subtropic high-controlled dry area, turning the dry-hot climate into a humid-hot climate and forming the most widely distributed Mesozoic coals in Asia (Boucot et al., 2009 and references therein) (Fig. 13). This could happen by the expanded Paleo-Asian continent and rising of many high mountains, such as the Paleo-Hoh Xil, Kunlun-Qinling and Indochina mountains from the Indochina Movement, mostly in the Late Triassic, which would have increased the heat contrast between Pangea (especially

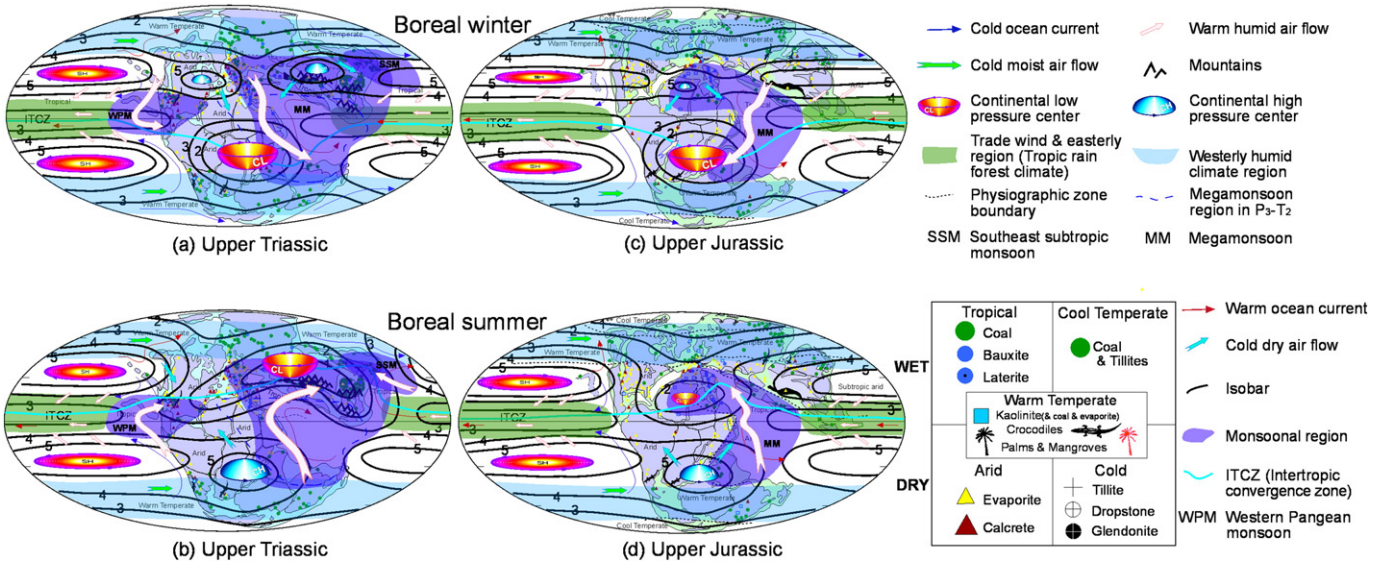


Fig. 13. Paleozoic–Mesozoic movement of the global and Qiangtang Blocks, climatic records and change and their coupling (note the Qiangtang Block is marked in black. Global blocks and climatic records come from Boucot et al. (2009) except for the Qiangtang Block where the data refer to this study). Note the megamonsoon region in the Late Triassic was remarkably expanded northeastwards compared to that in the Late Permian to Middle Triassic (maps not attached). Atmospheric isobar is relative without units.

the expanded Laurasia) and the Paleo-Tethys and increased the overall heat amount due to the high mountains receiving more sensible and latent heating. The expanded continent and increased heating in the southeastern margin of Laurasia caused northeastward migration or expansion of the Laurasian heating center in summer, thus driving the enhanced megamonsoon northeastward and expanding into the former dry Asian area (Fig. 13). Eastward expansion of the Pangean interior dry climate, a coupled or counterpart climate with a summer megamonsoon, to its eastern margin (Boucot et al., 2009) (Fig. 13a) lends support to our inference. This shift may have caused a trans-equator monsoon in the tropic western Pangean continent, which we called the western Pangean tropic monsoon (WPM), such as the present northern African tropic monsoon (Fig. 11), which would be responsible for the regional tropic humid and semi-humid climates in the tropic western Pangean continent (Fig. 13a, b). A possible weak southeast subtropic monsoon (SSM) could be formed as well in the southwestern rim of the northern subtropic high and northeastern Pangean continent, which could have caused a humid climate (Fig. 13b).

From the Late Jurassic, the QB climate turned into a dry climate and was intensified in the Cretaceous and Early Cenozoic (Fig. 2). We interpreted this as a result of the retreat and then termination of the megamonsoon, which was replaced by a subtropic high in response to the initial and eventually fast and complete dismission of the Pangean continent (Boucot et al., 2009) (Fig. 13c, d).

8. Interaction of the climate and basin tectonics and its controlling role on salt mineral formation

The sequences of the lithology and facies through the Permian to Cretaceous in different parts of the QB are a record of the interaction of the climate and basin tectonics with time (Fig. 2). As demonstrated above, the first-order climatic change in the QB from the Paleozoic to Mesozoic was determined by the coupling of the movements (positions) of the global and QB continents with the earth's heating zonation. The formation of the Pangea supercontinent broke the earth's heating zonation and generated a megamonsoon that had a profound impact on the global climate, including the QB, whereas breakup of Pangea terminated the megamonsoon, accompanied by the return of the earth's heating zonation, which controlled the climates of the continents under it. Moreover, successive breakup of the northeastern margin of the Gondwana continent formed sequential pieces of small-micro continent belts that moved northward and finally collided with Paleo-Asia. The successive breakups and consequent collisions determined the type, tectonics, deposition and deformation of small-micro continent-based basins. The breakup of the Gondwana continental margin created a rift basin that continuously opened as an ocean basin, such as the Paleo- or Meso-Tethys, in the southern margin of the block, where an association of rift- or ocean-volcanic and upper-fining sediment sequences formed, indicating an extensional depression and a late passive margin tectonic setting. On contrast collision of the block with Paleo-Asia would form a foreland basin along the collision zone, where an upper-coarsening sequence of sediments, typically from the flysch of a semi-deep ocean (slope) through the shallow sea-near shore sediments to continental molasse, suggesting a compressional depression-thrusting/folding tectonic setting. Therefore, the good coupling of an extremely dry climate with the tectonic basin (rift or foreland) would provide the greatest potential for the formation of salt and potash mineral sources.

Matching of the sequences of the lithology and facies through the Permian to the Cretaceous in different parts of the QB (Fig. 2) with the tectonics of the QB results in two time intervals, the Middle Triassic and the Late Jurassic, for the best coupling of dry climate with limited depression basin, which favors salt and potash formation. In the Middle Triassic, the QB had drifted into the north subtropic high (Figs. 12 and 13) and thick gypsum layers were found in the Middle Triassic red beds. Collision of the SQB with the NQB and closure of the LSSZ in the Middle Triassic (Yan et al., 2016) led to a limited E–W narrow depression, called the Tumengela Depression, that formed along the southern margin of the east CUB, and an N-stretched strait channel, called the Shuanghu Strait, that formed between the west and east CUB, which connected and provided sea water to the Tumengela Depression. The SQB was exposed without sedimentation, and the NQB was an open shallow sea deepening northwards into the Jinsha Ocean (HJSZ) (Fig. 14A). Therefore, if the connecting site between the Tumengela Depression and the Shuanghu Strait could be partially blocked, it would greatly favor salt and potash formation in the Tumengela Depression. The occurrence of thick gypsum layers along the southern margin of the CUB demonstrates the potential for salt and potash formation in the depression center.

In the Late Jurassic, the QB returned to a dry climate under the north subtropic high (Fig. 13c, d). After the Late Triassic, the closure of the HJSZ caused the NQB to almost close, with only the Shuanghu Strait open and connected to the SQB shallow Sea, which deepened southwards into the Bangong-Nujiang Ocean (BNSZ). The Late Jurassic–Early Cretaceous westward oblique closure of the BNSZ (Sengör et al., 1987; Yin and Harrison, 2000; Kapp et al., 2007; Zhu et al., 2011, 2013; Yan et al., 2016) caused sea water to retreat westwards from the QB. Tectonic upheaval was very important to isolate the eastern and central NQB for potential formation of salt and potash minerals because the global sea began a long rise, which has continued since the Late Jurassic (Fig. 2a). The occurrence of large lagoons (Fig. 14B) and the wide distribution of lagoon gypsum in the central and eastern NQB in the Late Jurassic stratigraphy (upper Xiali Fm. and Xueshan Fm.) (Wang et al., 2009) (Figs. 2–4) shed light on the potential formation of salts and potash in the NQB center. The gigantic salt and potash sources (up to 700×10^9 T potash) in the Karakum Basin in Tajikistan and Uzbekistan just to the west of the Qiangtang–Pamir Plateau were mainly formed in the Late Jurassic, with some lasting into the Early Cretaceous (Qian et al., 1994). The tectonics and climate coupling responsible for the source formation resembled the Qiangtang Basin, i.e., also due to the coupling of the closure of the Meso-Tethys (thus sea retreat westward from the Karakum Basin) with the subtropic dry climate. These results provide strong hints for the Qiangtang salt and potash formation.

However, subsequent tectonic compressions and folding–faulting, especially those in the Cenozoic due to collision of India with Asia, strongly faulted and folded and raised the stratigraphy, giving rise to strong erosion of the Xueshan Fm., which reduced the potential to conserve salt minerals.

9. Conclusions

(1) Detailed paleomagnetism with rock magnetism, constrained by fossils, indicated that the measured Yanshiping section was formed approximately >171.2–<157.5 Ma (late Aalenian to Oxfordian in the Middle to Late Jurassic) and the Quem Co, Buqu, Xiali, Suowa and Xueshan Fms. were formed approximately >171.2–165.5 Ma (Late Aalenian to early Callovian), 165.5–163.3 Ma (Middle to Late Callovian), 163.3–160.1 Ma (early Oxfordian), and 160.1–<157.5 Ma (late Oxfordian), with the top unmeasured YSP section lasting to the early stage of the Kimmeridgian (157.3–152.1 Ma).

(2) The lithological sequences in the Qiangtang Basin revealed sharp climatic changes from hot-humid Late Permian–Early Triassic, via hot-dry Middle Triassic and hot-humid Late Triassic–Middle Jurassic, to hot-dry Late Jurassic. Based on the proposed working model and combined lithological records in China and other parts of the world, we interpreted the Qiangtang Mesozoic climatic changes to have recorded the onset, off-monsoon and subtropic zone, intensified and eastward extension, and retreat/end of the megamonsoon. The changes demonstrated that the interactions between continental drift, mountain building, earth's heating

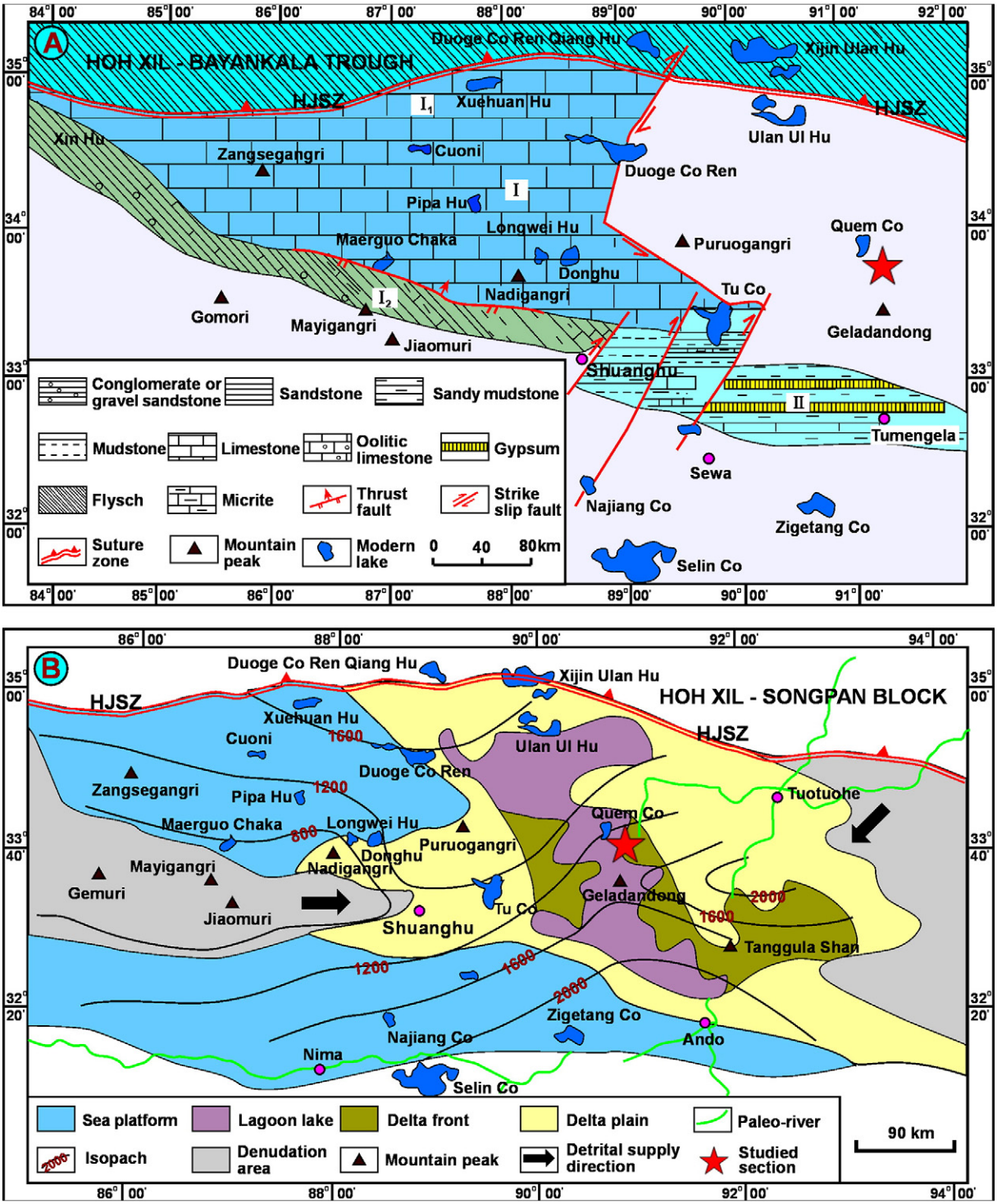


Fig. 14. Sedimentary facies distribution map of the Middle Triassic (A) and Late Jurassic (Xueshan Fm.) (B) in the Qiangtang Basin, showing the potential for salt and potash formation in the Tumengela Depression (Middle Triassic) and NQB (Late Jurassic) (modified from Wang et al., 2002; He et al., 2011). Note the NQB epicontinental sea retreated westwards. I, Sedimentary areas of the epeiric sea of North Qiangtang; I_1, Lithofacies associations of the shell carbonate rocks of Zangsegangri-Donghu; I_2, Lithofacies associations of the continent marginal siliciclast and carbonate rocks of Xinhu-Shuanghu; and II, Lithofacies associations of Tumengela Bay of South Qiangtang.

zonation and megamonsoon evolution were the most influential mechanisms driving the Qiangtang Basin climatic changes.

(3) The Late Jurassic coupling between the tectonic rising of the Qiangtang Basin due to closure of the Meso-Tethys and the dry climate due to the retreat/end of the megamonsoon and return of the subtropic high reached the most favorable conditions for the formation of evaporite and possible potash mineral sources in the basin.

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